



An industrial process optimization framework: from data to deployment with case studies in food production processes

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Abstract

Control systems are crucial for ensuring operational efficiency, safety, and product consistency in industrial environments. Industrial AI, which integrates artificial intelligence technologies into traditional control systems, offers significant performance enhancement opportunities. However, current approaches face two fundamental challenges: theoretical research often develops advanced AI methods disconnected from practical industrial needs, while industrial implementations tend toward fragmented, case-specific solutions (addressing issues in isolation without systematic approaches), hindering standardization and scalability. To bridge these gaps, we propose an industrial AI development framework designed to systematically enhance system performance through integrated methodologies encompassing data acquisition and analysis, model development, and system deployment. The framework's effectiveness and generalizability were validated through applications in two industrial processes: steam flow management in wort kettles and filtration process optimization. Empirical testing demonstrated the framework's generalizability across different control paradigms, achieving substantial performance improvements compared to the conventional control system. Beyond operational improvements, this framework could enable substantial energy efficiency gains and carbon footprint reduction through optimized process control, addressing growing sustainability imperatives in energy-intensive industrial operations.

Keywords Industrial process · Data science · Artificial intelligence · Optimization framework · Control systems · Food production processes

Introduction

Manufacturing constitutes a cornerstone of the global economy, with value added and output in the global manufacturing market projected to amount to \$14.34 trillion and \$50.84 trillion in 2025, respectively (Statista, 2024). Industrial automation has transformed these operations from simple mechanical devices to sophisticated computer-controlled systems (Nambiar et al., 2023), with single-variable control loops maintaining process measurements at desired set points (Fig. 1). Recent studies indicate that 78% of manufacturers have implemented or plan to invest in advanced planning software (Ajiga et al., 2024).

The Fourth Industrial Revolution (Industry 4.0) leverages technologies such as Industrial Internet of Things (IIoT), artificial intelligence (AI), and data analytics to create smart manufacturing systems (Qiu et al., 2025). The ISA-95 architecture (Fig. 2) illustrates how advanced process controls enable system-wide communication, facilitating real-time data analysis and innovations such as digital twins and predictive maintenance (Javaid et al., 2022; Malik et al., 2021).

Despite this potential, the realization of Industry 4.0 faces challenges with complex, nonlinear processes that often operate in isolation, lacking integration with enterprise-level systems (Ahmed et al., 2022; Rajagopal et al., 2022; Yang & Ge, 2022). Theoretical research typically develops algorithms without considering industrial constraints like robustness and interpretability, while industrial implementations adopt fragmented solutions addressing isolated problems (Espina-Romero et al., 2024; Windmann et al., 2024).

To address these challenges, this paper proposes a framework for industrial AI development that systematically enhances system performance. The framework comprises

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Fig. 1 Traditional industrial automation systems: a single process control targets a given set point

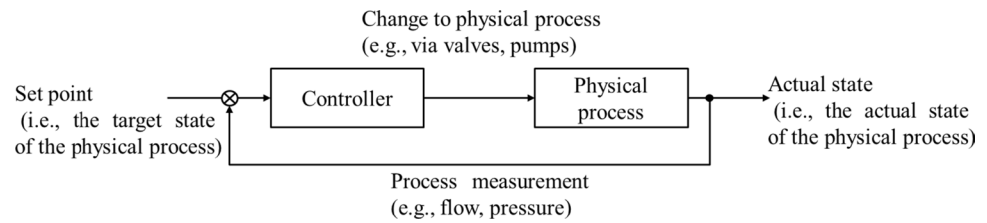
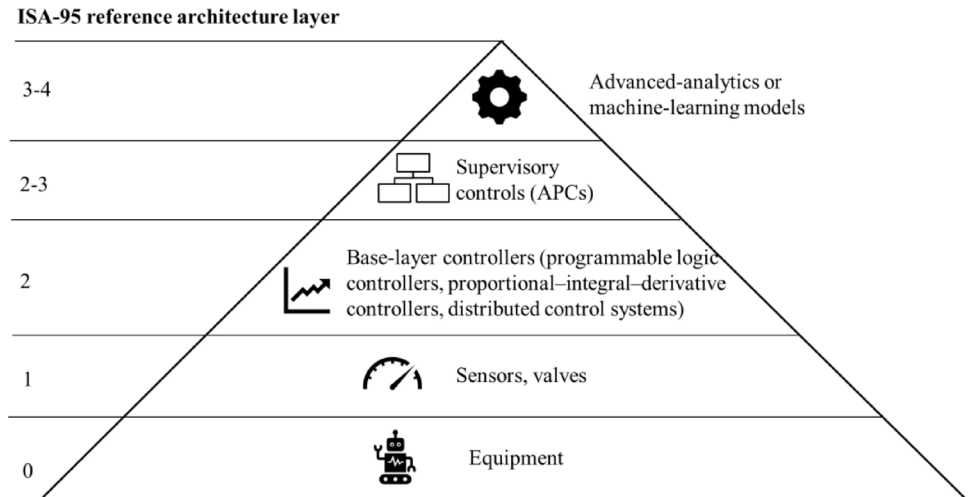


Fig. 2 ISA-95 reference architecture for advanced process controls (APCs) in manufacturing (Görner et al., 2020)



a five-stage workflow: *Scenario Understanding, Problem Identification, Optimization Strategy Formulation, Optimization Strategy Deployment, and Verification and Evaluation* (Kaymakci et al., 2021). This structure bridges theoretical advancements with practical implementation requirements, establishing clear procedures for data acquisition, model development, and deployment. The framework was validated through two case studies in food production processes, demonstrating substantial improvements over traditional methods (Jagatheesaperumal et al., 2021). This application illustrates how the framework identifies system characteristics, develops appropriate control strategies, and deploys solutions enhancing energy efficiency and product quality. This research makes three principal contributions:

1. To bridge the theory-practice gap in industrial AI development, we develop a five-stage framework providing end-to-end guidance from problem identification to verified industrial operation.
2. To address nonlinear valve dynamics under variable steam pressure, we propose pressure-dependent gain scheduling achieving substantial overshoot reduction without requiring complex process models.
3. To establish framework generalizability, we demonstrate effectiveness across two contrasting control paradigms (continuous setpoint tracking vs. discrete batch optimization) with quantified performance improvements.

The remainder of this paper is organized as follows: section “[Literature review](#)” reviews the current state of industrial AI research and implementation, identifying critical gaps in existing methodologies. Section “[An industrial AI framework for optimizing process controls](#)” presents the proposed framework with detailed workflow descriptions. Sections “[Case study 1: steam flow management](#)” and “[Case study 2: filtering process optimization](#)” provide comprehensive validations through two case studies: steam flow management and filtration process optimization. Section “[Discussion](#)” discusses framework generalizability, limitations, and practical implications. Section “[Conclusion](#)” concludes with key findings and future research directions.

Literature review

The literature review examines the current state of research and implementation of artificial intelligence in industrial process control, focusing on the advances, challenges, and opportunities in both theoretical and industrial contexts. This analysis provides a foundation for understanding the proposed industrial AI framework and its potential contributions to addressing existing gaps between theoretical research and practical applications.

Theoretical research in industrial AI

Theoretical research in industrial AI has expanded dramatically with Industry 4.0, creating a diverse ecosystem of intelligent industrial systems. This expansion integrates artificial intelligence with traditional manufacturing processes within the Fourth Industrial Revolution framework (Peres et al., 2020). Researchers have developed specialized AI techniques addressing manufacturing challenges through system optimization, predictive maintenance, and adaptive control methodologies (Kim et al., 2022). The theoretical foundation includes intelligent control theory applying computational methods to enhance manufacturing efficiency and product quality (Li et al., 2017), addressing requirements for operational flexibility and process resilience.

Machine learning technologies constitute the foundational layer of modern industrial AI research, with approaches tailored to different manufacturing challenges. The research encompasses supervised learning for predictive models, unsupervised learning for pattern recognition, and reinforcement learning for optimizing control parameters (Carpanzano & Knüttel, 2022; Lui et al., 2024). Deep learning architectures have demonstrated particular efficacy in processing complex industrial data from sensors, vision systems, and time-series databases (Plathottam et al., 2023), enabling advanced manufacturing functions including automated inspection and process optimization.

Industrial AI applications have evolved from isolated implementations to comprehensive frameworks spanning entire manufacturing value chains. Research extends AI capabilities across product design, process control, quality assurance, and supply chain optimization (Lee, 2023). Hybrid methodologies combining physics-based models with data-driven techniques have emerged as particularly valuable, addressing limitations of purely empirical models where limited data constrains conventional machine learning approaches (Wang et al., 2022).

Integration frameworks provide essential architectural blueprints for implementing AI within established industrial infrastructures. Yang et al. (2021) established reference architectures integrating AI with industrial internet platforms, while Lavin et al. (2022) contributed Technology Readiness Level frameworks for industrial machine learning deployment. These frameworks address the challenge of translating theoretical capabilities into practical implementations, establishing pathways for technology integration across diverse manufacturing settings (Zheng et al., 2018).

However, translating industrial AI from laboratory research to operational implementation faces persistent challenges related to practical deployment constraints. Many promising solutions remain confined to controlled environments that inadequately represent the complexity of actual

industrial operations (Plathottam et al., 2023). Research often prioritizes algorithmic refinement over implementation considerations such as integration with legacy systems and operational robustness (Tyagi et al., 2024). This gap is widened by insufficient evaluation methodologies (Jan et al., 2023) and limited focus on model interpretability, which industrial stakeholders identify as prerequisites for adoption (Elahi et al., 2023).

Recent research initiatives have increasingly addressed implementation challenges by incorporating industrial constraints directly into AI development methodologies. Zhou et al. (2012) demonstrated this through intelligent control systems engineered for industrial processes, while Carletti et al. (2019) focused on enhancing interpretability of manufacturing models. Sinha and Lee (2024) discussed the challenges and strategies for effectively integrating AI into industrial systems, focusing on maximizing benefits and addressing technical, ethical, and regulatory concerns. These efforts indicate a growing recognition within the theoretical community of the need to bridge the gap between theoretical advancements and practical industrial requirements, although significant challenges remain in translating research into standardized, deployable solutions.

Industrial implementation and applications

Industrial applications of AI prioritize practical implementation and measurable business outcomes rather than theoretical advancement. Industrial practitioners focus on solutions that integrate with existing infrastructure, deliver tangible benefits, and operate reliably under real-world conditions. Successful implementations demonstrate significant improvements in operational efficiency, product quality, and cost reduction. For example, Siemens has implemented AI-driven predictive maintenance systems analyzing production data to identify potential equipment failures before occurrence, resulting in reduced downtime and maintenance costs (Abdul Jamil et al., 2025).

Industrial AI adoption typically follows a targeted use-case approach addressing specific operational challenges rather than implementing comprehensive frameworks (Alexopoulos et al., 2020). According to ISG research, over 90% of manufacturers implementing AI are motivated by business growth, innovation, and efficiency improvements (Baljevic, 2024). Despite these motivations, AI adoption in manufacturing lags behind other industries due to information technology (IT)—operational technology (OT) convergence gaps, data quality issues, and difficulties capturing domain knowledge for machine learning purposes (Baljevic, 2024). These implementation challenges reflect the complex realities of industrial environments, where

legacy systems and operational constraints significantly complicate AI deployment.

Real-world industrial AI implementations demonstrate both potential benefits and practical implementation hurdles. Many companies have invested in Advanced Process Controls (APCs), yet often utilize only a fraction of their potential value, with some installations operating at less than 10% of capacity due to insufficient tuning and maintenance. Companies like BASF have successfully deployed AI solutions across chemical manufacturing processes, including reaction optimization, predictive maintenance, and quality monitoring, but these implementations required significant customization to adapt AI solutions to specific industrial contexts.

The implementation gap is further evidenced by the “pilot purgatory” phenomenon, where AI initiatives remain in experimental phases without transitioning to operational deployment. A primary factor contributing to this challenge

is the lack of standardized reference architectures for industrial AI that would provide frameworks for scaling applications across manufacturing enterprises (Baljevic, 2024). Additional obstacles include workforce skill deficits, organizational change management challenges, and integration complexities with existing industrial control systems (Kutz et al., 2022). IBM highlights operational risks associated with AI model reliability and security concerns when integrating AI into industrial control environments (Stryker, 2025).

The disparity between theoretical research and industrial implementation represents a fundamental barrier to advancing intelligent process control in manufacturing. While theoretical research focuses on algorithm development and theoretical performance improvements, industrial applications require robust, scalable, and interpretable solutions operating within manufacturing constraints (Sinha & Lee, 2024). This disconnect has resulted in fragmented, case-specific implementations addressing isolated problems rather than providing comprehensive frameworks for AI integration across manufacturing operations. Bridging this gap requires approaches combining theoretical rigor with practical industrial considerations, emphasizing standardization, scalability, and integration with existing manufacturing infrastructure (Jan et al., 2023).

Research gap and framework limitations

Despite the extensive research in industrial AI, existing methodologies exhibit significant limitations that hinder effective implementation in manufacturing process control environments (as shown in Table 1). This analysis reveals critical gaps in current approaches.

Current methodologies suffer from a fundamental theory-practice disconnect, where ISA-95 provides enterprise architecture without AI-specific guidance, while CRISP-DM variants focus on data analytics without addressing industrial process control constraints. Enhanced CRISP-DM for Manufacturing acknowledges operational challenges but lacks systematic scenario understanding and industrial-specific problem identification (Bokrantz et al., 2024). These approaches promote fragmented, case-specific solutions contributing to “pilot purgatory,” where AI initiatives remain experimental without operational deployment. Additionally, existing frameworks inadequately address integration challenges with operational technology infrastructure and lack systematic performance validation methodologies for industrial applications (Plathottam et al., 2023).

The literature reveals a critical need for a systematic framework that bridges theoretical AI capabilities with practical industrial constraints, provides comprehensive guidance from problem identification through deployment

Table 1 Comparative analysis of existing industrial AI framework

Framework	Performance characteristics	Scalability	Deployment ease	Key limitations
ISA-95 (Görner et al., 2020)	No AI-specific methodology; High complexity	High complexity limits scalability; Requires complete system restructuring	Extensive infrastructure changes	No AI-specific guidance for process control
CRISP-DM (Huber et al., 2019)	Generic approach; Limited industrial constraints	Cross-industry applicability but poor industrial integration	Poor integration with factory equipment	Not designed for real-time control
Enhanced CRISP-DM (Bokrantz et al., 2024)	Data-centric focus; Limited deployment framework	Manufacturing-specific but limited process diversity	Insufficient hardware integration guidance	Data-centric without control theory integration
DMME (Huber et al., 2019)	Narrow engineering scope; Limited optimization	Engineering-specific applications only	Inadequate for comprehensive AI deployment	Limited to data acquisition phase only
Proposed framework	Validated across multiple control paradigms	Modular design enables cross-process application	Edge-computing architecture for legacy system integration	Systematic methodology from problem identification to deployment

verification, integrates with existing operational technology infrastructure, and offers quantitative performance validation specific to industrial process control applications. As demonstrated in Table 1, the proposed framework uniquely addresses these requirements by providing end-to-end methodology—from problem identification to operational deployment—validated across multiple control paradigms, while existing approaches either focus solely on enterprise integration (ISA-95) or data analytics (CRISP-DM variants) without systematic guidance for industrial AI implementation.

An industrial AI framework for optimizing process controls

To address the identified limitations in existing methodologies—particularly the theory-practice disconnect, fragmented implementations, and inadequate integration with operational technology infrastructure—we propose a comprehensive industrial AI framework specifically designed for process control optimization. Unlike current approaches that focus on enterprise-level integration (ISA-95) or generic data analytics (CRISP-DM variants), this framework provides systematic guidance from problem identification through deployment verification, with demonstrated effectiveness across diverse industrial applications.

The framework follows a structured, iterative workflow comprising five essential stages that guide industrial AI implementation from conceptualization to operation, as shown in Fig. 3. This systematic approach begins with scenario understanding, where system structures and key performance indicators are clearly defined through on-site investigation. The workflow then advances to problem identification, where historical operational data is analyzed to clarify system characteristics and define appropriate control objectives. Following this, improvement strategy formulation employs physics-based, data-driven, or hybrid methodologies to develop mathematical models and generate corresponding algorithm code that address specific control problems. The process continues with system deployment, integrating hardware and software components within existing infrastructure to enable real-time operation. Finally, verification and evaluation quantitatively assess performance improvements through statistical comparison of pre- and post-implementation metrics. This methodical workflow

provides practitioners with a practical roadmap for transforming conventional control systems into data-driven solutions tailored to address specific industrial challenges.

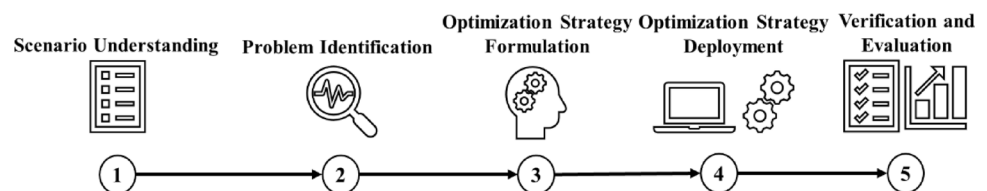
Development workflow

Scenario understanding

Initiating the development of industrial AI requires a comprehensive understanding of the system characteristics and components. This process begins with on-site investigations to collect information regarding structural components, operational principles, and control mechanisms, alongside defining key performance indicators as optimization targets:

- (1) *Process flow investigation* entails analyzing the fundamental operational mechanisms within the system, including process flow diagrams that delineate operation sequences, inputs, and outputs. This examination clarifies cause-and-effect relationships governing system behavior, documenting process flow diagrams, energy pathways, key parameters, and interrelationships between process stages.
- (2) *System structure clarification* involves cataloging physical components including actuators and sensors, alongside their spatial configuration. This comprehensive mapping identifies critical components influencing system performance, documenting hardware specifications, communication protocols, and network architectures facilitating data exchange and control.
- (3) *Worker operating modes* focuses on understanding human–system interactions, including standard procedures for machine configuration and parameter adjustment. The investigation documents interface mechanisms, decision-making protocols for setpoint adjustments, and tacit knowledge influencing system performance.
- (4) *Objective specification* identifies intended performance targets, which may include minimizing energy consumption, maximizing throughput, or ensuring process stability. These measurable objectives align with operational requirements and guide subsequent problem identification and control system modeling phases.

Fig. 3 An industrial process optimization framework



Problem identification

This stage focuses on analyzing historical operational data to identify system characteristics and control challenges through three interconnected processes.

- (1) *Data Collection and Preprocessing* involves gathering operational data from sensors and control systems, followed by exploratory analysis using visualization techniques. The data undergoes quality assessment, cleaning procedures, and outlier detection, with feature engineering performed to derive additional variables capturing system dynamics.
- (2) *System Understanding* compares sensor data with documented processes to verify alignment between theoretical and actual behavior. Parameter curves for critical devices are analyzed to evaluate performance metrics, identifying degradation patterns or operational anomalies.
- (3) *Optimization Opportunity Identification* analyzes performance plots to identify significant deviations from desired parameters. Key variables with strong correlations to performance metrics are prioritized for control strategies, with root cause analysis tracing performance issues to fundamental sources.

This analytical approach transforms historical data into actionable insights regarding system behavior, constraints, and optimization opportunities, providing the foundation for subsequent control system modeling and algorithm development.

Optimization strategy formulation

Following problem identification, this stage focuses on determining appropriate improvement strategies, formulating mathematical relationships, and developing implementation code. The primary objective is establishing effective strategies based on identified system characteristics and performance requirements.

Central to this process is precisely formulating the control problem through mathematical expressions. This begins by defining three key variable categories: measurement variables providing feedback on system states, control variables directly manipulated to influence behavior, and decision variables determining setpoints based on control algorithms.

The final component involves translating mathematical formulations into executable implementation code. This process converts improvement strategies into practical algorithms deployable on industrial control hardware,

addressing computational efficiency, integration with existing systems, and robustness against operational variations.

Optimization strategy deployment

The improvement strategy deployment phase bridges the gap between theoretical algorithm development and practical industrial implementation. With deployable algorithm code prepared, this stage focuses on establishing physical and digital infrastructure for operational integration.

Hardware integration establishes computing infrastructure, typically employing edge devices positioned within the industrial network to interface with existing controllers. Communication protocols provide data exchange mechanisms, implemented as function libraries enabling algorithms to receive sensor data and transmit control signals using standards such as Modbus or OPC UA. Software implementation creates the runtime environment with multi-threading capabilities, error handling, and logging functionalities.

These elements function as a cohesive system, with hardware providing computational foundation, communication protocols facilitating data flow, and software orchestrating algorithm execution. The deployment concludes with commissioning activities verifying proper integration and performance under operational conditions.

Verification and evaluation

The verification and evaluation stage quantifies the impact of deployed intelligent process control through systematic comparison of pre- and post-implementation data. Verification confirms system feasibility in the industrial environment, examining reliability, stability, and interaction with existing control systems across various production scenarios.

Performance evaluation employs statistical comparison methodologies to measure improvements in key performance indicators. This analysis examines metrics such as control accuracy, energy efficiency, and process stability, applying appropriate statistical techniques to determine the significance of observed changes.

System deployment architecture

This section provides a potential system deployment architecture, as shown in Fig. 4. It is an ecosystem spanning cloud resources, local control systems, and monitoring interfaces. This architecture establishes bidirectional information flow between operational technology and analytical capabilities for intelligent process control implementation.

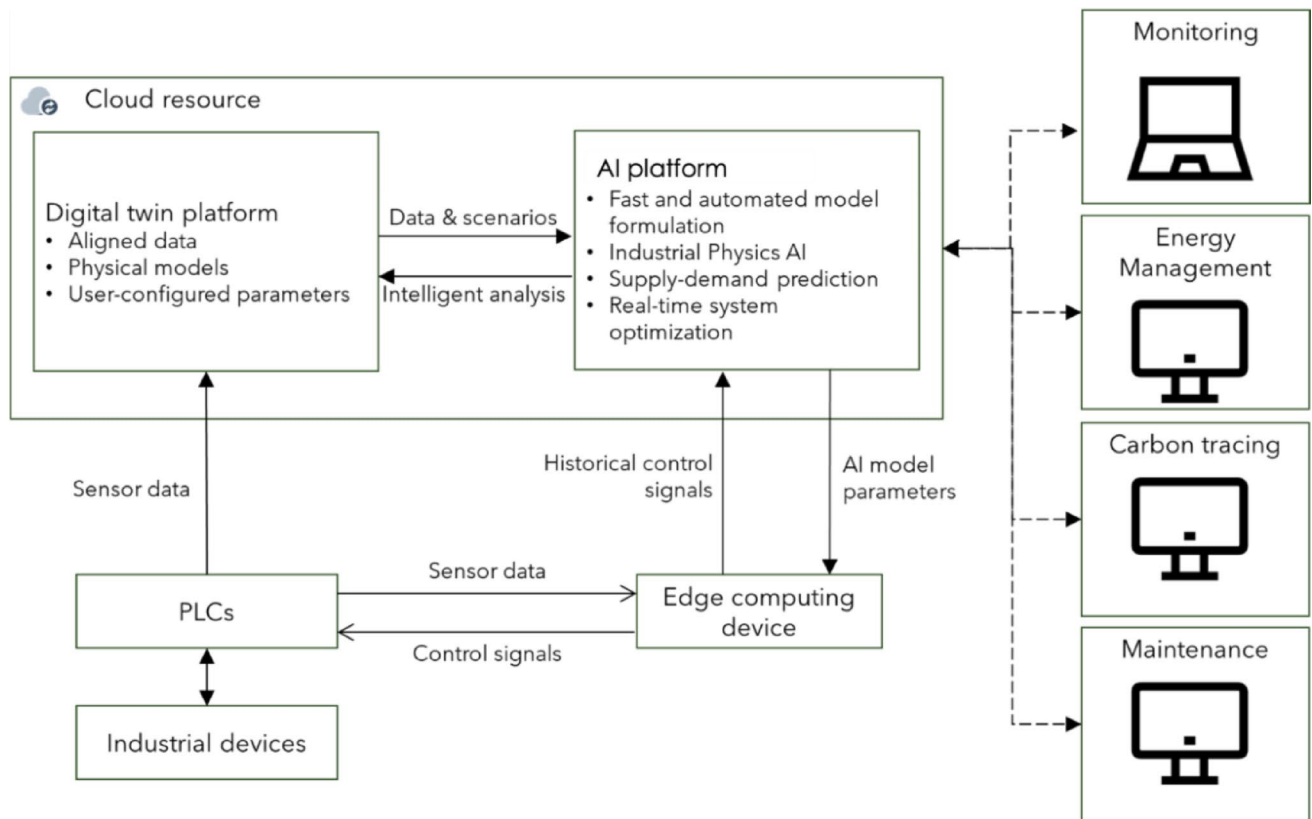


Fig. 4 An example of intelligent process control system deployment architecture

The cloud resource layer consists of a digital twin platform and an AI platform. The digital twin platform creates virtual representations with aligned data structures and physical models, while the AI platform performs advanced modeling using industrial physics AI, supply-demand prediction, and real-time optimization algorithms. The local control layer comprises Programmable Logic Controllers (PLCs) interfacing with industrial devices and edge computing devices implementing AI algorithms. PLCs collect sensor data from equipment and transmit this information to both cloud resources and edge devices. Edge computing devices execute control algorithms locally using AI model parameters developed in the cloud environment. Monitoring and management interfaces extend functionality through visualization capabilities for system monitoring, energy management, carbon emissions tracking, and maintenance planning. These interfaces leverage integrated system data for operational oversight and decision-making.

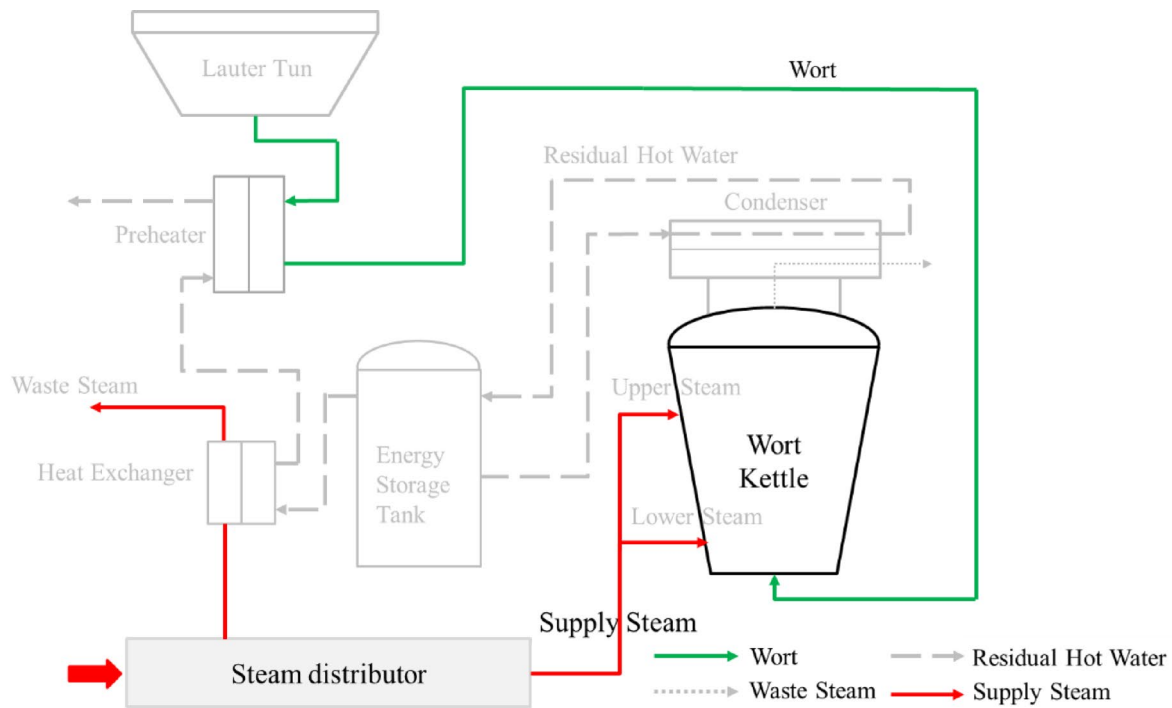
This deployment architecture establishes a continuous information cycle where sensor data flows to cloud resources and edge devices, the cloud environment refines AI models, edge devices implement these models to generate optimized control signals, and PLCs execute these signals while collecting new data, creating a self-improving system that balances innovation with operational stability.

Case study 1: steam flow management

Scenario understanding: steam flow management

(1) Process flow investigation

Beer production process encompasses multiple critical stages including malting, mashing, lautering, preheating, boiling, fermentation, and packaging (Mainardis et al., 2024). Among these, the boiling stage merits particular attention due to its dual impact on product quality and energy consumption (Pino et al., 2024). Figure 5a illustrates the flow of wort from the lauter tun into the wort kettle during the boiling stage, with steam supplied from the steam distributor providing thermal energy to the kettle. Figure 5b, c depict the wort kettle system and the steam valve, respectively. This stage constitutes the most energy-intensive component of the production process, accounting for 35 to 50% of the total energy consumption throughout the production cycle. The boiling process is essential for optimal extraction of flavor compounds from hops, directly influencing the beer's sensory characteristics (Shopska et al., 2022). In the case study, our research focuses on steam flow management during the boiling stage.



(a) Energy and material flow diagram of the boiling stage



(b) A wort kettle system(Eresson, 2025)



(c) Steam valves of a wort kettle(TLV, 2025)

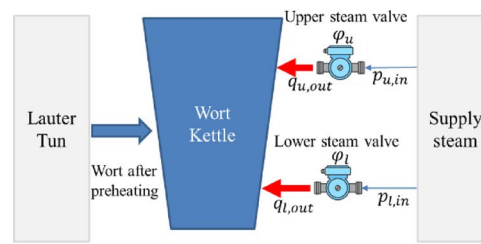
Fig. 5 Process flow and devices of the boiling stage. **a** Energy and material flow diagram of the boiling stage, **b** a wort kettle system (Eresson, 2025), **c** steam valves of a wort kettle (TLV, 2025)

The facility utilizes three 800-HL wort kettles, each equipped with dual steam valves for flow regulation (shown in Fig. 6a). The steam flow dynamics are governed by the relationship between steam valve opening $\phi_j, j \in \{u, l\}$ and supply steam pressure $p_{j,out}, j \in \{u, l\}$, with the steam flow rate $q_{j,in}^a, j \in \{u, l\}$ determined by both factors. The subscripts u and l denote the upper and lower steam valves, respectively. Table 2 presents the primary symbols and notation utilized in this analysis.

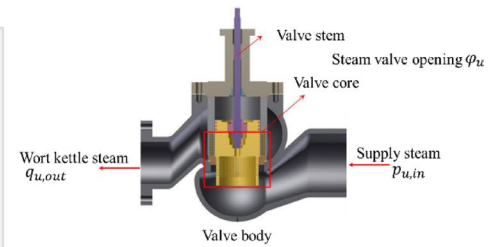
Wort kettles include steam valves, valve actuators, steam pressure sensors, and flow rate transducers. Each wort kettle is equipped with dual steam valves (designated as $j \in \{u, l\}$) that function as the critical actuators controlling the boiling process. As depicted in Fig. 6b, these steam valves consist of a valve body, stem, and core, with the latter serving as the key component for regulating steam valve opening and controlling steam flow rate (Chen & Jin, 2021).

The boiling stage begins with preheated wort being transferred to wort kettles, where high-temperature steam initiates

Fig. 6 Schematic representations. **a** A wort kettle system, **b** a steam valve, (Chen & Jin, 2021)



(a) A wort kettle system



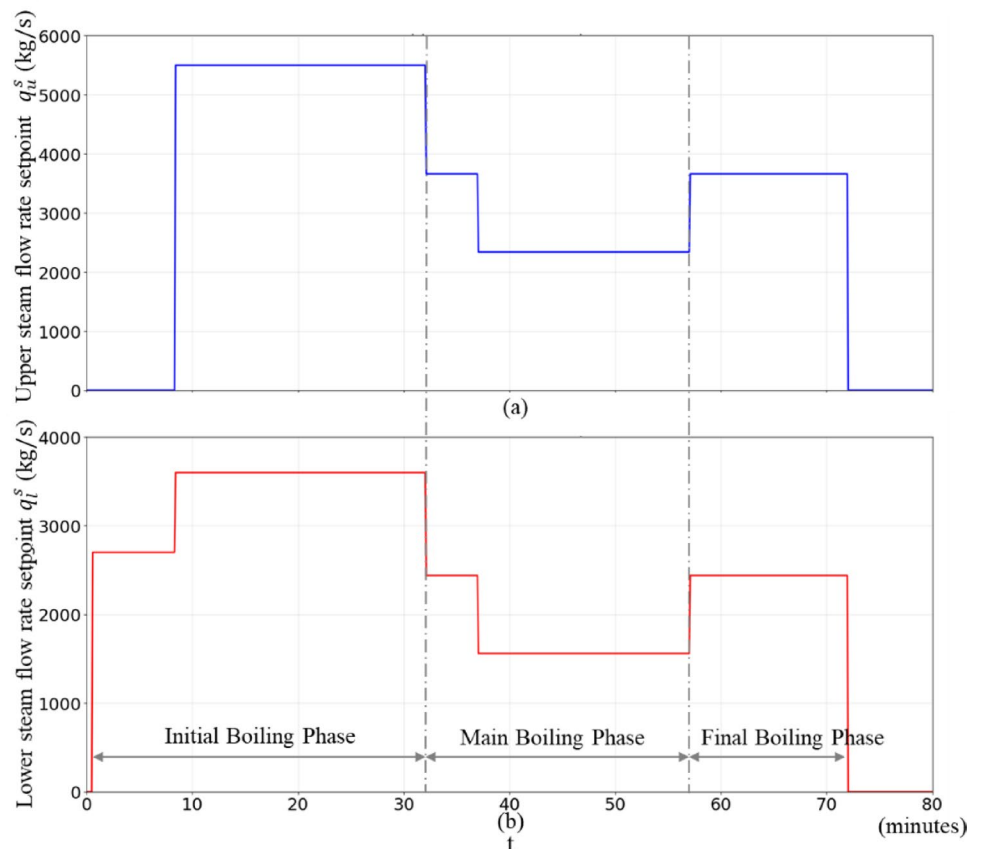
(b) A steam valve (Chen & Jin, 2021)

Table 2 Definitions of symbols

Symbol	Definition
$q_{j,out}(t)$	The steam flow rate of steam valve j at time t
$q_{j,out}^a(t)$	The actual steam flow rate of steam valve j at time t
$q_{j,out}^s(t)$	The setpoint steam flow rate of steam valve j at time t
$\phi_j(t)$	The steam valve opening of steam valve j at time t
$p_{j,in}(t)$	The supply steam pressure of steam valve j at time t
$e_j(t)$	The steam flow rate tracking error of steam valve j at time t
ϵ_1	Maximum allowable steam flow rate overshoot
ϵ_2	Maximum allowable steam flow rate undershoot
j	Index for the steam valve $j \in \{u, l\}$
t	Continuous time variable over the control horizon $t \in [0, T]$

and maintains the boiling process (Oyewola et al., 2024). Figure 7 illustrates the steam flow rate profiles customized for the specific requirements of the brewing process. The complete boiling cycle comprises three distinct phases, each characterized by different steam flow rate setpoints. Initial Boiling Phase: High steam flow rates enable rapid heating of wort to boiling temperature. During this phase, wort continues to flow into the kettle. The operational sequence involves activating the lower steam valve first, followed by the upper steam valve once the liquid level reaches a predetermined threshold. Main Boiling Phase: Steam flow rates are reduced to maintain stable boiling while optimizing energy consumption. This phase ensures consistent temperature maintenance with minimal energy expenditure. Final Boiling Phase: A moderate increase in steam flow rates facilitates final flavor development, followed by rapid

Fig. 7 Profiles of steam flow rate



reduction to zero at process completion. Precise management of steam flow rates throughout these phases ensures both process effectiveness and final product quality.

The steam flow rate profiles are implemented through an integrated Human-Machine Interface (HMI) that provides comprehensive visualization and control capabilities. The HMI presents operators with real-time graphical representations of setpoint curves alongside actual measured steam flow rates for both upper and lower valves, enabling instantaneous identification of deviations between desired and actual performance.

The control architecture follows a hierarchical structure integrating the HMI with the plant's Distributed Control System (DCS) infrastructure. Setpoint values are stored in the DCS database and transmitted to individual PLCs governing each wort kettle via Ethernet communication protocols. The PLCs process these commands and generate control signals for steam valve actuators. Sensor feedback data returns through the reverse pathway to update the HMI displays, establishing a closed-loop control system. This architecture ensures that theoretical steam flow profiles translate into precise physical control actions while maintaining real-time operational visibility. The HMI incorporates recipe management capabilities, allowing operators to modify steam flow profiles for different beer varieties while preserving process consistency and quality standards.

The primary objective of steam flow management is to maintain the actual steam flow rate in accordance with the setpoint values. Figure 8 illustrates the current control architecture, which comprises a PID controller for each steam valve (upper and lower valves). The PID controller generates a control signal $\phi_j(t)$ based on the deviation between the desired setpoint $q_{j,out}^s(t)$ and the actual steam flow rate

$q_{j,out}^a(t)$, where $j \in \{u, l\}$ denotes the upper and lower valves, respectively. The steam valve responds to this control signal while being influenced by supply steam pressure $p_{j,in}(t)$.

(2) System structure clarification

The system incorporates positioned measurement devices to ensure precise control feedback, as illustrated in Fig. 8. Steam flow rate sensors are installed downstream of each steam valve at a distance of 2–3 pipe diameters from the valve outlet, a configuration designed to minimize turbulence-induced measurement errors. Supply steam pressure transducers are mounted upstream of the valves to provide stable pressure readings unaffected by valve operation dynamics. Valve position sensors are integrated directly into the actuator assemblies to deliver real-time feedback on valve opening percentage.

This sensor configuration enables accurate measurement of the three critical control variables: actual steam flow rate $q_{j,out}^a(t)$, supply steam pressure $p_{j,in}(t)$, and valve opening $\phi_j(t)$. The downstream flow measurement ensures detection of actual delivered steam rather than theoretical flow, while upstream pressure sensing captures supply-side variations that influence valve characteristics. These sensors transmit digital signals to PLCs via MQTT protocol, which serve as intermediate data aggregation nodes before transmitting information to the centralized DCS.

(3) Worker operating modes

The facility features automated boiling operations with remote management capabilities. Operators control the

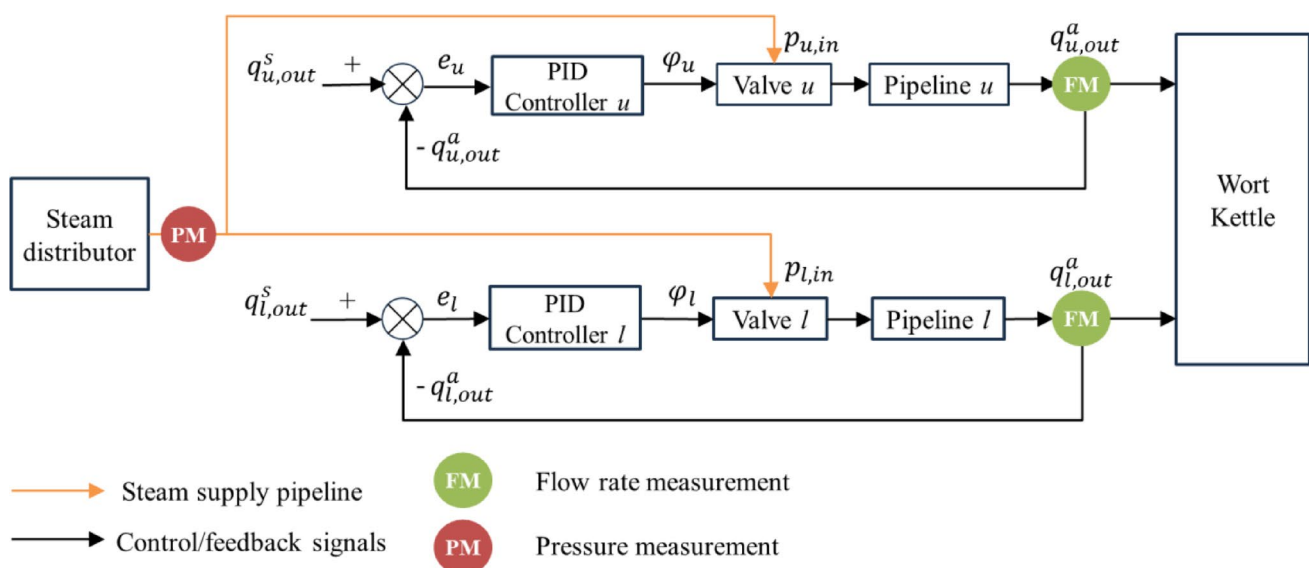


Fig. 8 Current control architecture for steam flow management (non-AI)

system through an HMI that provides real-time monitoring of steam flow rates, valve openings, and supply steam pressures, with measurements updating every 6–8 s. The DCS provides comprehensive visualization of operational parameters for all relevant equipment.

(4) Objective specification

Due to process requirements, steam flow rate serves as a critical parameter affecting the boiling process. Consequently, the primary objective of this project is to optimize steam flow management to enhance both the consistency of product quality and energy efficiency.

Problem Identification

(1) Data collection and preprocessing

Historical operational data were extracted from the facility's DCS, encompassing continuous measurements of steam flow rates, steam valve opening, and supply steam pressures at 6–8 s intervals. The dataset comprises approximately 7 million data points collected over a 7-month operational period (September 2024 to March 2025), with each month contributing approximately 1 million data points. Data sampling intervals of 6–8 s across three wort kettles yield approximately 7200 measurements per 12-hour production cycle for each wort kettle, providing extensive temporal coverage across seasonal variations and diverse production scenarios. Primary variables and performance metrics monitored during this study are summarized in Table 3. Standard data preprocessing protocols were implemented to ensure data integrity.

Table 3 Primary variables and performance metrics

Category	Parameter	Range/deviation
Primary	Steam flow rate $q_{j,out}^a(t)$	0–8000 kg/s
Variables	Supply steam pressure $p_{j,in}(t)$	3.0–5.0 kgf/cm ²
	Steam valve opening $\phi_j(t)$	0–100% of maximum
Performance	Overshoot of steam flow rate O_{steam}	$\leq 25\%$ of setpoint value
Metrics	Steam flow rate deviation E_{cum}	± 2500 kg/s from setpoints
	Steam usage deviation D_{steam}	± 1000 kg from standard values
	Total evaporation rate deviation D_{evap}	$\pm 2\%$ from standard values
	Total boiling time deviation D_{boil}	± 15 min from standard process time

(3) System understanding

Taking the upper steam valve as an example, an analysis of actual versus setpoint steam flow rates during the boiling stage revealed control inadequacies in the system. The setpoint profile, as illustrated in Fig. 9a, was established by the brewery according to specific production process requirements. In this project, this setpoint profile remains fixed and predetermined. Supply steam pressure is dependent on supply-side pressure conditions and cannot be controlled in this project. This pressure parameter influences steam flow rates, with steam valve opening serving as the only controllable variable within the system.

As illustrated in Fig. 9a, the system exhibited deviations following setpoint modifications, with overshoot reaching 18.2% and undershoot extending to -10.9% . These deviations indicated limitations in the control system's ability to maintain stable flow rates under varying operational conditions.

Figure 9b displayed the corresponding real-time valve opening ϕ_u . Examination of valve opening ϕ_u showed a correlation with flow rate $q_{u,out}^a$, indicating that excessive valve adjustments contribute to observed flow instability.

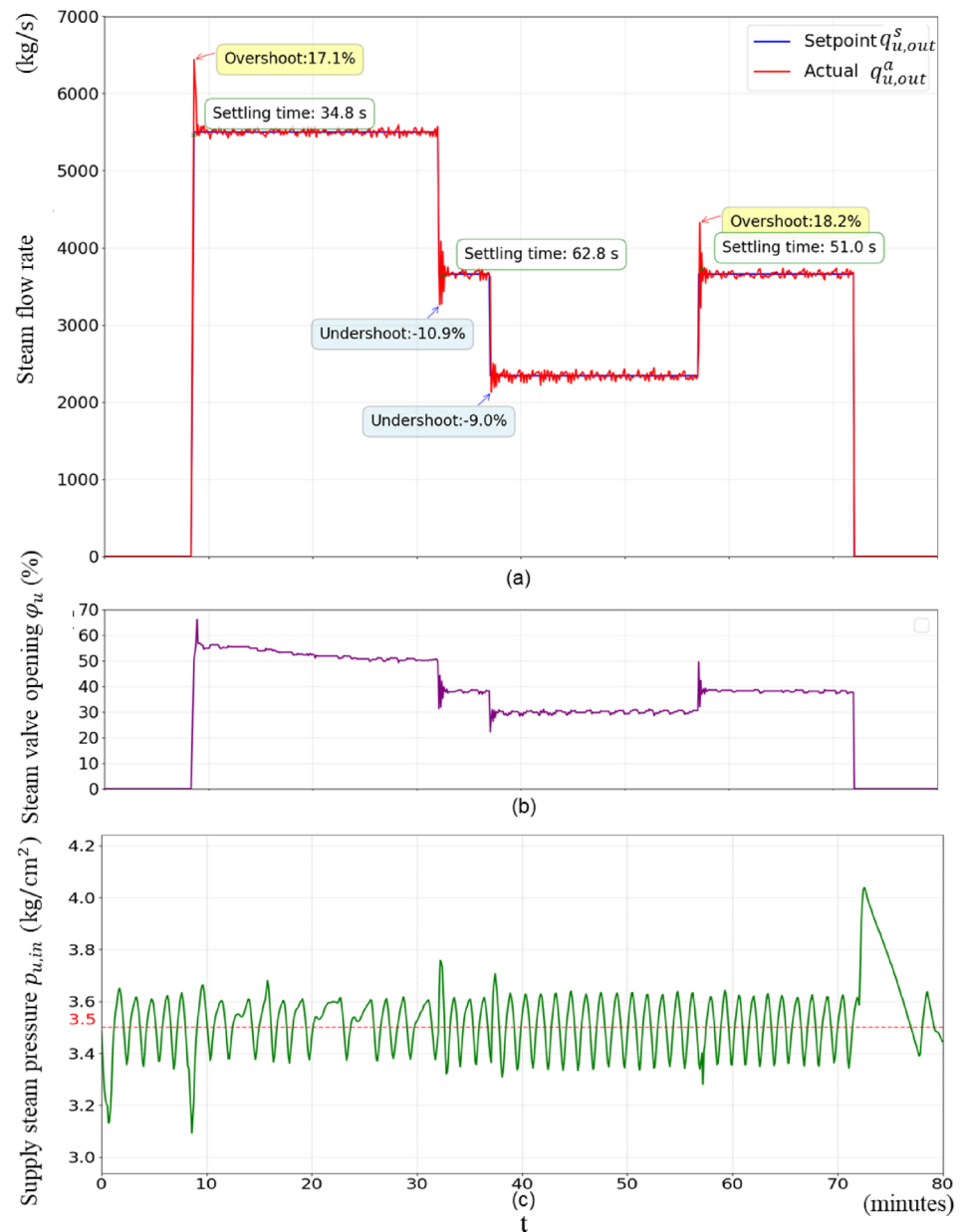
Further analysis of supply steam pressure $p_{u,in}$ identified temporal fluctuations from the nominal value of 3.5 kgf/cm², as shown in Fig. 9c. These pressure variations, attributable to concurrent operation of other steam-consuming equipment and source pressure irregularities, introduce additional complexity to effective flow management.

The fluctuations in supply steam pressure illustrated in Fig. 9c suggest interactions between supply steam pressure and steam flow management. To elucidate the system dynamics, Fig. 10 characterizes the relationship between steam flow rate $q_{u,out}^a$ and valve opening ϕ_u across four discrete supply steam pressure ranges: 3.0–3.5, 3.5–4.0, 4.0–4.5, and 4.5–5.0 kgf/cm². The data demonstrate a correlation between valve opening and flow rate within each pressure band, with gradient increasing at elevated pressures. For example, at 50% valve opening, the flow rate at 4.5–5.0 kgf/cm² exceeds that at 3.0–3.5 kgf/cm². This pressure-dependent relationship underscores the complexity of the control problem, as higher supply steam pressures amplify the flow response to steam valve opening adjustments.

(3) Optimization opportunity identification

Based on the comprehensive analysis of historical operational data, the primary limitation in the current steam flow management approach was identified. The specific control objective is to minimize tracking error between setpoint and actual steam flow rates while constraining overshoot and

Fig. 9 Steam flow rate, steam valve opening, and supply steam pressure over time



undershoot within predefined thresholds, thereby ensuring both energy efficiency and product quality consistency.

Optimization strategy formulation

Optimization model

The nonlinear relationship between steam valve opening $\phi_j(t)$, supply steam pressure $p_{j,in}(t)$, and steam flow rate $q_{j,out}^a(t)$ reveals that system gain varies across different pressure ranges, indicating that a single set of PID parameters cannot deliver optimal performance across the full spectrum of operating conditions encountered in daily production. Root cause analysis traced performance deviations

to this fundamental system characteristic, identifying supply steam pressure as the critical variable with substantial influence on control response. Therefore, the influence of supply steam pressure should be considered in the steam flow management process.

To address these challenges, an adaptive PID control strategy was proposed that adjusts controller parameters based on supply steam pressure. As shown in Fig. 11, this approach preserves the PID structure while introducing adaptability to accommodate supply steam pressure $p_{j,in}(t)$ fluctuations. By implementing pressure-dependent gain scheduling, the controller can adjust its response characteristics to maintain consistent performance regardless of supply steam pressure conditions. The adaptive mechanism modifies controller

Fig. 10 Relationship between $q_{u,out}$ and ϕ_u across different supply steam pressure ranges

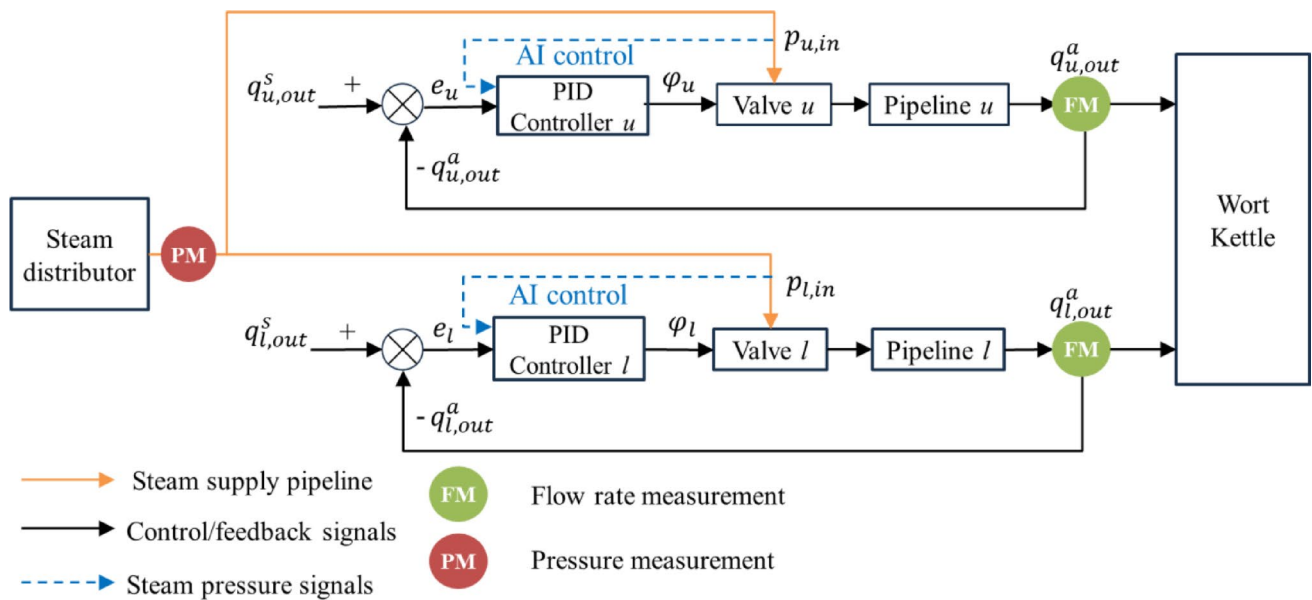
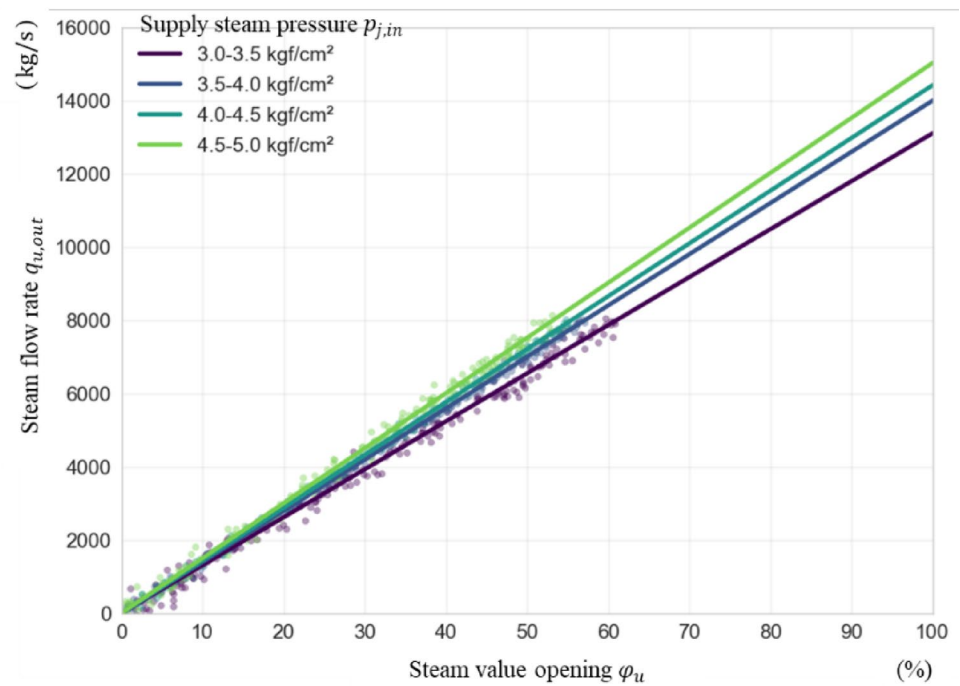


Fig. 11 AI control-based feedback loop for steam flow management

gains according to measured supply steam pressure, thereby compensating for the nonlinear pressure-flow relationship. This approach can be formulated as an optimization problem to determine the optimal gain adjustment parameters $(\theta_{p,j}, \theta_{i,j}, \theta_{d,j})$ that minimize steam flow rate tracking error across the entire operating range.

The following section presents a mathematical formulation of this adaptive control strategy, establishing the objective function, constraints, and parameter adjustment mechanisms necessary to achieve precise steam flow

management under varying supply steam pressure conditions. For steam valve $j \in \{1, 2\}$, the optimization problem is formulated as

$$\min_{\theta_{p,j}, \theta_{i,j}, \theta_{d,j}} J_j = \int_0^T |q_{j,out}^s(t) - q_{j,out}^a(t)| dt, \quad (1)$$

$$\text{Subject to} \quad \max(0, q_{j,out}^a(t) - q_{j,out}^s(t)) \leq \epsilon_1 \cdot q_{j,out}^s(t), \quad \forall t \in [0, T], \quad (2)$$

$$\max(0, q_{j,out}^s(t) - q_{j,out}^a(t)) \leq \epsilon_2 \cdot q_{j,out}^s(t), \quad \forall t \in [0, T], \quad (3)$$

$$\phi_j(t) = K_{p,j}(t)e_j(t) + K_{i,j}(t) \int_0^t e_j(\tau) d\tau + K_{d,j}(t) \frac{de_j(t)}{dt}, \quad (4)$$

$$e_j(t) = q_{j,out}^s(t) - q_{j,out}^a(t), \quad (5)$$

$$p_j^{norm}(t) = \frac{p_{j,in}(t)}{p_{in}^s}, \quad (6)$$

$$K_{p,j}(t) = \theta_{p,j,0} + \theta_{p,j,1} \cdot (1 - p_j^{norm}(t)), \quad (7)$$

$$K_{i,j}(t) = \theta_{i,j,0} + \theta_{i,j,1} \cdot (1 - p_j^{norm}(t)), \quad (8)$$

$$K_{d,j}(t) = \theta_{d,j,0} + \theta_{d,j,1} \cdot (1 - p_j^{norm}(t)), \quad (9)$$

$$q_{j,out}^a(t) = G_j(\phi_j(t), p_{j,in}(t)), \quad \forall t \in [0, T], \quad (10)$$

$$0 \leq \phi_j(t) \leq \phi_j^{max}, \quad \forall t \in [0, T], \quad (11)$$

$$q_{j,out}^a(t) \geq 0, \quad \forall t \in [0, T]. \quad (12)$$

$$0 \leq K_{p,j}(t), K_{i,j}(t), K_{d,j}(t) \leq 10, \quad \forall t \in [0, T]. \quad (13)$$

The objective Eq. (1) is to minimize the L_1 -norm between the setpoint steam flow rate $q_{j,out}^s(t)$ and the actual steam flow rate $q_{j,out}^a(t)$ over the control horizon $[0, T]$, which ensures tracking performance. Additionally, other performance metrics—including steam usage deviation, total evaporation rate deviation, and total boiling time deviation—are fundamentally dependent on the deviation between actual steam flow rate and its setpoint, making them indirectly affected by this control objective.

Overshoot constraint: Equation (2) limits the extent to which the actual steam flow rate can exceed the setpoint by a maximum value of ϵ_1 . This constraint prevents overpressure risks and ensures system safety.

Undershoot constraint: Equation (3) ensures the actual flow rate does not fall below the setpoint by more than ϵ_2 , maintaining process stability and consistent steam supply.

Adaptive PID control law: Equation (4) defines the control input $\phi_j(t)$ as generated by a PID controller with time-varying gains. This adaptive approach addresses limitations

in traditional PID controllers that do not account for fluctuations in supply steam pressure.

Error definition: Equation (5) specifies the steam flow rate tracking error $e_j(t)$ as the difference between setpoint and actual flow rates, which serves as input to the controller.

Normalized supply steam pressure: Equation (6) defines the normalized pressure $p_{j,in}^{norm}(t)$ as the ratio of the actual supply steam pressure $p_{j,in}(t)$ to the setpoint value p_{in}^s . A ratio greater than 1 indicates that the current pressure exceeds the setpoint, and vice versa.

Gain adjustment mechanism: Equations (7)–(9) define how the PID gains ($K_{p,j}(t)$, $K_{i,j}(t)$, $K_{d,j}(t)$) adapt based on supply steam pressure $p_{j,in}(t)$. For example, when supply steam pressure is low, proportional gain increases to reduce error quickly; when pressure is high, gain decreases to prevent overshooting. Each gain comprises a base component and a pressure-dependent variable component.

- $\theta_{p,j,0}$ is the basic proportional gain.
- $\theta_{i,j,0}$ is the basic integral gain for eliminating steady-state bias.
- $\theta_{d,j,0}$ is the basic derivative gain for enhancing stability and reducing overshoot.
- The variable components $\theta_{p,j,1} \cdot (1 - p_j^{norm}(t))$, $\theta_{i,j,1} \cdot (1 - p_j^{norm}(t))$, and $\theta_{d,j,1} \cdot (1 - p_j^{norm}(t))$ adjust gains according to normalized supply steam pressure variations.
- The comprehensive gain parameters are thus formulated as: $\theta_{p,j} = (\theta_{p,j,0}, \theta_{p,j,1})$, $\theta_{i,j} = (\theta_{i,j,0}, \theta_{i,j,1})$, $\theta_{d,j} = (\theta_{d,j,0}, \theta_{d,j,1})$

System dynamics: Equation (10) relates the actual steam flow rate to valve opening and supply steam pressure through function G_j .

Control input bounds: Equation (11) restricts valve opening to physical limitations, where ϕ_j^{max} represents the maximum possible valve opening.

Non-Negativity of flow: Equation (12) enforces the physical constraint that flow rates cannot be negative.

PID gain bounds: Equation (13) ensures time-varying gains remain within practical operational limits to maintain system stability.

Optimization algorithms

To effectively solve the optimization problem formulated in Eqs. (1)–(13), a Particle Swarm Optimization (PSO) algorithm was implemented for determining the optimal gain adjustment parameters $\theta_{p,j}, \theta_{i,j}, \theta_{d,j}$. PSO was selected due to its efficacy in handling non-linear, high-dimensional optimization problems without requiring gradient information, which is particularly advantageous for complex control systems with potentially discontinuous search spaces.

Valve characteristic model estimation To facilitate the optimization process, an accurate mathematical representation of the relationship between valve opening, supply steam pressure, and steam flow rate was essential. Based on fluid dynamic principles and empirical observations from Fig. 10, a valve characteristic model was formulated:

$$q_{j,out}^a(t) = G_j(\phi_j(t), p_{j,in}(t)) = \frac{\phi_j(t)}{100} \cdot q_{j,out}^{max} \cdot \sqrt{\frac{p_{j,in}(t)}{p^{ref}}}, \quad (14)$$

where $q_{j,out}^a(t)$ is the actual steam flow rate at time t , $\phi_j(t)$ is the steam valve opening, $q_{j,out}^{max}$ is the maximum flow rate of the valve, $p_{j,in}(t)$ is the current supply steam pressure, p^{ref} is the reference pressure (3.5 kgf/cm²).

This model captures the essential nonlinear relationship between valve opening, pressure, and flow rate, with flow proportional to valve opening and square root of normalized pressure, consistent with standard valve flow characteristics observed in industrial applications.

PSO algorithm implementation The PSO algorithm employs a population of candidate solutions (particles) that navigate the search space to identify optimal parameters. Each particle represents a complete solution vector $\theta_j = (\theta_{p,j,0}, \theta_{p,j,1}, \theta_{i,j,0}, \theta_{i,j,1}, \theta_{d,j,0}, \theta_{d,j,1})$ contain-

ing all six adaptive PID controller parameters for steam valve j . The algorithm operates as follows:

Initialization: A swarm of N_p particles is randomly generated within the parameter bounds, with each particle assigned an initial velocity vector.

$$X_i^0 = [x_{i,1}^0, x_{i,2}^0, \dots, x_{i,6}^0], \quad i = 1, 2, \dots, N_p$$

$$V_i^0 = [v_{i,1}^0, v_{i,2}^0, \dots, v_{i,6}^0], \quad i = 1, 2, \dots, N_p$$

where X_i^0 represents the initial position (parameter values) of particle i , and V_i^0 denotes its initial velocity.

Discretization of the continuous-time optimization problem: The numerical implementation of the continuous-time optimization problem requires discretization, where the objective function becomes.

$$J_j = \sum_{k=0}^N |q_{j,out}^s(t_k) - q_{j,out}^a(t_k)| \Delta t, \quad (15)$$

with the discretized control input:

$$\begin{aligned} \varphi_j(t_k) = & K_{p,j}(t_k) e_j(t_k) \\ & + K_{i,j}(t_k) \sum_{n=0}^k e_j(n) \Delta t + K_{d,j}(t_k) \frac{e_j(t_k) - e_j(t_{k-1})}{\Delta t}. \end{aligned} \quad (16)$$

For this implementation, the control horizon T corresponds to the total boiling time for each production batch (typically 90 min in this brewing process). The discretization uses $N = 3600$ steps based on the control system's update frequency of 1.5 s, enabling responsive adaptation to changing process conditions throughout the entire boiling stage. This formulation enables practical implementation of the adaptive control strategy in digital control systems.

Fitness evaluation: Each particle's fitness is evaluated by simulating the closed-loop system response using the particle's parameter values and calculating the objective function.

$$f(X_i) = \sum_{k=0}^N |q_{j,out}^s(t_k) - q_{j,out}^a(t_k)| \Delta t,$$

subject to the constraints defined in Eqs. (2)–(13). A penalty function approach is employed to handle constraint violations, particularly for the overshoot and undershoot constraints in Eqs. (2)–(3):

$$f_{\text{penalty}}(X_i) = f(X_i) + \lambda_1 \sum_{k=0}^N \max(0, q_{j,\text{out}}^a(t_k) - q_{j,\text{out}}^s(t_k) - \epsilon_1 \cdot q_{j,\text{out}}^s(t)) + \lambda_2 \sum_{k=0}^N \max(0, q_{j,\text{out}}^s(t_k) - q_{j,\text{out}}^a(t_k) - \epsilon_2 \cdot q_{j,\text{out}}^a(t)),$$

where λ_1 and λ_2 are penalty coefficients.

Position update: In each iteration t , particles update their positions based on their velocity vectors.

$$X_i^{t+1} = X_i^t + V_i^{t+1}.$$

The velocity update incorporates both cognitive and social components:

$$V_i^{t+1} = \omega V_i^t + c_1 r_1 (P_i - X_i^t) + c_2 r_2 (G - X_i^t),$$

where:

- ω is the inertia weight that balances global and local search capabilities,
- c_1 and c_2 are cognitive and social acceleration coefficients,
- r_1 and r_2 are random numbers in the range $[0,1]$,
- P_i is the best position found by particle i (personal best),
- G is the best position found by any particle (global best).

Boundary handling: To ensure parameter values remain within operational constraints, a reflective boundary condition is implemented.

If $x_{i,d}^{t+1} < x_{\min,d}$, then $x_{i,d}^{t+1} = x_{\min,d}$ and $v_{i,d}^{t+1} = -v_{i,d}^{t+1}$,

If $x_{i,d}^{t+1} > x_{\max,d}$, then $x_{i,d}^{t+1} = x_{\max,d}$ and $v_{i,d}^{t+1} = -v_{i,d}^{t+1}$,

where $x_{\min,d}$ and $x_{\max,d}$ represent the lower and upper bounds for parameter dimension d .

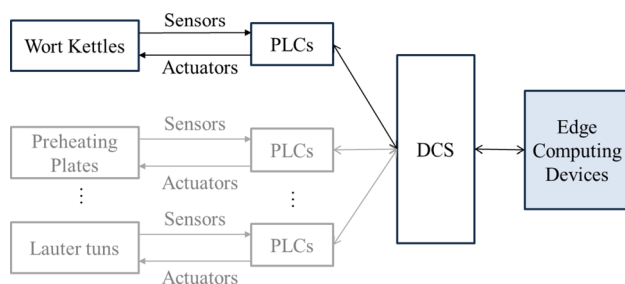


Fig. 12 Deployed edge computing-based hierarchical control system architecture

Termination criteria: The algorithm terminates when either a maximum number of iterations $T_{\max}$ is reached or the global best solution remains unchanged for a pre-defined number of consecutive iterations T_{conv} , indicating convergence.

System deployment

The implementation of process control for wort kettles required deployment of edge computing devices to establish the computational infrastructure for operational integration. These edge computing devices were deployed within the control framework to handle multiple functions: acquiring sensor signals from the DCS, transmitting control signals back to the DCS, and executing specialized control algorithms for the wort kettles across the entire production system. Figure 12 illustrates the communication architecture among edge computing devices, DCS and PLCs.

The data acquisition and control architecture operates as follows: Sensors and actuators connected to process equipment (wort kettles, preheating plates, etc.) communicate with their respective PLCs via MQTT protocol. These PLCs function as intermediate data aggregation nodes, collecting sensor measurements and executing local control functions before transmitting consolidated information to the centralized DCS. This hierarchical arrangement facilitates monitoring and control of multiple process units within the production facility.

The DCS processes and relays operational data to the edge computing devices, which serve as the primary computational nodes for control algorithm execution. Within this edge computing layer, the adaptive PID algorithm is implemented to generate optimized control signals based on real-time process measurements. These control signals, specifically the calculated steam valve opening, are transmitted through the reverse communication pathway (edge computing $\rightarrow$ DCS $\rightarrow$ PLCs $\rightarrow$ actuators) to execute precise control actions at the physical equipment level.

The control feedback loop continuously monitors critical process variables—specifically steam flow rate and supply steam pressure—enabling dynamic adjustment of valve openings to maintain the desired flow characteristics despite system disturbances. This closed-loop architecture provides robust performance in the presence of the pressure fluctuations previously identified in the problem analysis section. System commissioning procedures included comprehensive verification of bidirectional communication pathways, characterization of control response dynamics, and integration validation testing to ensure operational reliability under varying process conditions.

Verification and evaluation

Performance metrics

System performance was evaluated using five quantitative metrics designed to assess control precision, energy efficiency, and process consistency:

- (1) Overshoot of steam flow rate (O_{steam}): quantifies the maximum deviation between actual flow and setpoint

$$O_{steam} = \max_t |q_{j,out}^s(t) - q_{j,out}^a(t)| / q_{j,out}^s(t). \quad (17)$$

- (2) Steam Flow Rate Deviation (E_{cum}): Defined as the cumulative absolute error between the actual steam flow rate and setpoint, quantifying overall tracking performance.

$$E_{cum} = \sum_t^T |q_{j,out}^s(t) - q_{j,out}^a(t)| \Delta t, \quad (18)$$

where Δt represents the sampling time interval.

- (3) Steam usage deviation (D_{steam}): measures the difference between actual steam consumption and standardized reference values

$$D_{steam} = Q_{actual} - Q_{standard}, \quad (19)$$

where Q_{actual} represents measured steam usage, and $Q_{standard}$ represents reference steam consumption.

- (4) Total Evaporation Rate Deviation (D_{evap}): Expresses the percentage deviation in wort evaporation from established process standards.

$$D_{evap} = \frac{R_{actual} - R_{standard}}{R_{actual}} \times 100\%, \quad (20)$$

where R_{actual} and $R_{standard}$ represent the actual and reference evaporation rates, respectively.

- (5) Total boiling time deviation (D_{boil}): quantifies Temporal deviations in the boiling process

$$D_{boil} = T_{actual} - T_{standard}, \quad (21)$$

where T_{actual} and $T_{standard}$ represent the actual and standard boiling durations.

Open-loop testing

Following the mathematical formulation and implementation of the adaptive control strategy, validation of its performance characteristics was essential to confirm the theoretical advantages before full-scale deployment. To evaluate the fundamental response characteristics of both control systems, we conducted systematic open-loop testing to isolate controller behavior from feedback effects. Figure 13 presents representative response profiles from these trials, illustrating steam flow rate dynamics following identical setpoint changes under both non-AI control and AI control.

The non-AI control (Fig. 13a) exhibits significant control limitations when subjected to varying pressure conditions. When a step change in setpoint from 0 kg/h to 3800 kg/h was introduced at $t=57$ min, the system demonstrates overshoot of approximately 18.2%. In contrast, the AI control (Fig. 13b) demonstrates improved performance characteristics under identical test conditions. Following the same setpoint transition, maximum undershoot is limited to 5.7%.

The open-loop test results demonstrate that pressure-dependent adaptive gain scheduling provides an effective solution to the inherent limitations of fixed-parameter controllers when operating in variable pressure environments. The system's ability to dynamically adjust control parameters based on real-time supply steam pressure enables optimal response characteristics across the full operating range, addressing both principal challenges identified in the problem analysis: control response limitations and pressure-dependent system dynamics.

Figure 14 illustrates the optimized adaptive response of PID parameters to steam pressure variations. As supply steam pressure increases, the proportional gain $K_p(t)$ decreases to prevent excessive control action that could induce oscillations in high-pressure conditions. Conversely, at lower pressures, elevated proportional gain ensures sufficient controller responsiveness. The integral gain $K_i(t)$ demonstrates similar pressure-dependent adaptation, while derivative gain $K_d(t)$ exhibits more moderate variation to maintain stability across operating conditions.

This pressure-dependent parameterization enables the controller to compensate for the nonlinear relationship between steam valve opening and steam flow rate across different pressure regimes, thereby addressing the fundamental limitation identified in the problem analysis. The differential responses of gain parameters across the pressure spectrum confirm the effectiveness of the optimization approach in capturing the complex system dynamics and establishing appropriate control responses for each operating condition.

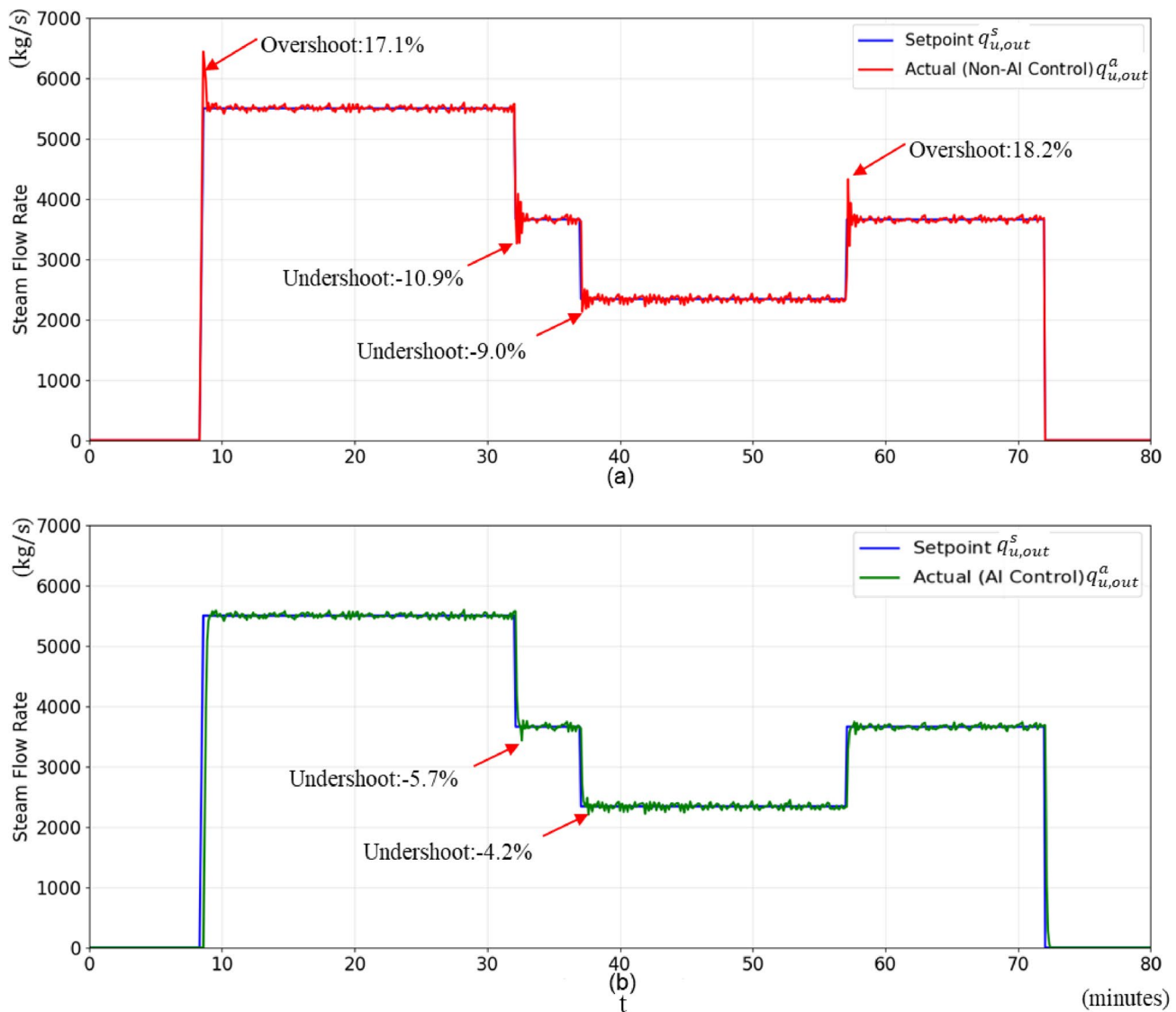


Fig. 13 Steam flow management under **a** non-AI control and **b** AI control

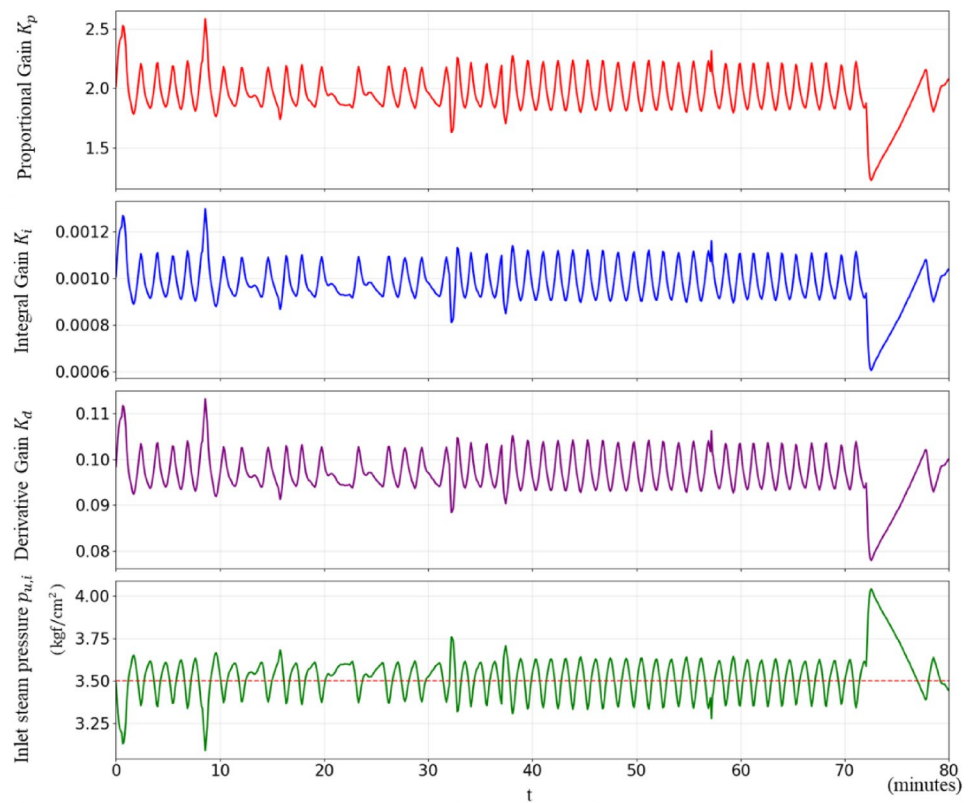
Closed-loop testing

Comprehensive performance analysis was conducted using production data from 82 batches under non-AI control and a single batch under AI-enhanced control during the period from January 1 to April 15, 2025. Statistical distribution analysis of performance metrics facilitated systematic comparison between control methodologies. Figure 15 presents probability density distributions of the five key performance indicators for a wort kettle under both control conditions.

The non-AI control system demonstrates limited capability to maintain stable process parameters, as evidenced by:

- (1) Reduction in overshoot distribution with a mean of 16.78% under non-AI control compared to 8.00% under AI control, resulting in energy inefficiency.
- (2) Higher mean steam flow rate deviations reaching 179.06 kg for the upper kettle and 87.05 kg for the lower kettle, while AI control values measured only 45.77 kg and 32.87 kg respectively, indicating persistent tracking errors.
- (3) Higher mean steam usage deviation measured at 71.04 kg compared to 25.42 kg under AI control, reflecting deviations from reference standards.
- (4) Similar evaporation rate deviation distributions between control systems with a mean of 0.85% under non-AI control compared with 0.88% under AI control, indicating comparable performance for this parameter.
- (5) Complex boiling time deviation pattern exhibiting a bimodal distribution under non-AI control.

Fig. 14 Adaptive PID parameters response to steam pressure changes



These observations confirm that non-AI control strategies exhibit limitations in managing the complex, multivariable dynamics of the boiling stage, particularly under conditions of fluctuating supply steam pressure. The observed process variability has direct implications for both energy utilization efficiency and final product quality consistency.

AI-driven steam flow control enhances boiler efficiency by stabilizing demand and reducing load fluctuations. Stable steam demand optimizes air-fuel ratios and combustion, minimizing inefficiencies from rapid load changes. Assuming that the stabilized steam demand achieved through AI control contributes to approximately 0.5–1% improvement in overall boiler thermal efficiency. For a typical brewery producing 1000 batches annually with a mean steam consumption of 2,500 kg per batch, the boiler efficiency improvement results in projected annual steam savings of 12,500–5000 kg. Considering industrial steam generation efficiency of approximately 80% and natural gas heating value of 50 MJ/kg, these efficiency improvements correspond to estimated energy savings of 781–1563 GJ annually.

Comparative analysis of optimization algorithms and control methods

(1) Comparison with other optimization algorithms

Subsequently, to validate PSO selection for parameter optimization, a comparative analysis was conducted against Genetic Algorithm (GA) and Simulated Annealing (SA) under identical test conditions. All algorithms optimized the adaptive PID parameters using the same objective function and constraints defined in Eqs. (1)–(13). The results are listed in Table 4.

PSO demonstrated superior performance across all metrics, achieving 8.3% reduction in overshoot compared to GA and 9.7% compared to SA. Similarly, undershoot was reduced by 3.5% and 7.1% respectively. Computational efficiency was significantly improved, with PSO converging 26.5% faster than GA and 3.9% faster than SA. The comparative evaluation confirms PSO's effectiveness for industrial process control optimization, delivering both superior control performance and computational efficiency compared to conventional optimization approaches.

(2) Advanced control methods benchmark comparison

To comprehensively validate the proposed adaptive PID approach, a comparison was conducted against applicable advanced control methods under identical steam flow management conditions. As shown in Table 5, the proposed adaptive PID demonstrated superior performance across metrics. Fuzzy Logic Control showed limitations in precision control, while Neural Network Control required

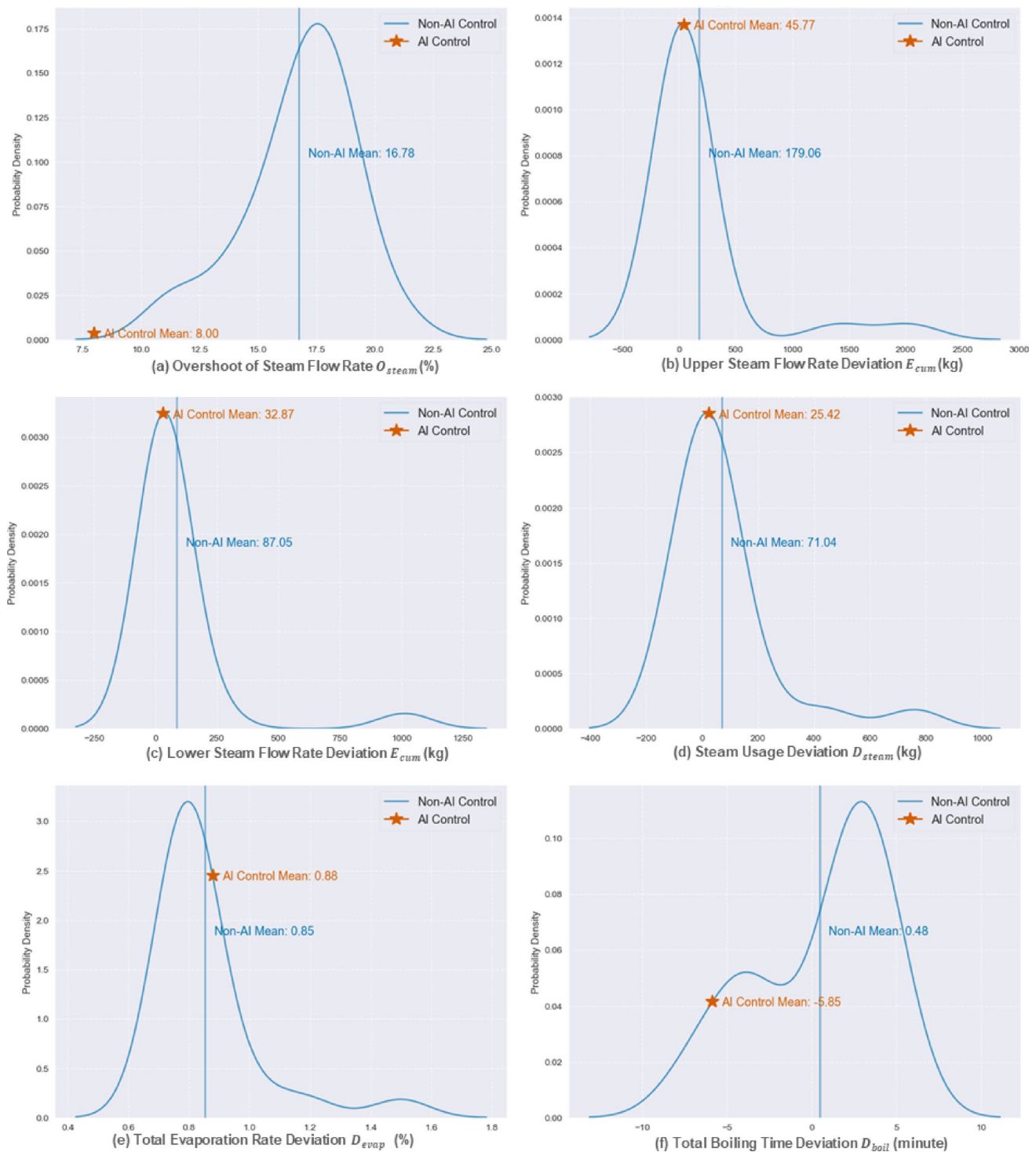


Fig. 15 Performance comparison of non-AI control and AI control in wort kettles

Table 4 Comparative performance of optimization algorithms

Performance metric	PSO	GA	SA
Maximum overshoot (%)	3.05	3.25	3.29
Maximum undershoot (%)	5.7	5.9	6.1
Steam flow rate deviation (kg)	30.5	31.2	32.5
Convergence time (s)	54.3	68.7	56.1

extensive training data and specialized maintenance expertise. The adaptive PID's model-free adaptation mechanism, computational efficiency, and seamless DCS integration make it optimal for this industrial application, demonstrating that control method selection should prioritize practical implementation requirements alongside performance

Table 5 Performance comparison of advanced control methods

Performance metric	Proposed adaptive PID	Fuzzy logic	Neural network
Maximum overshoot (%)	3.05	3.80	3.45
Maximum undershoot (%)	5.70	6.10	5.85
Steam flow rate deviation (kg)	30.5	35.8	31.3
Computational time (ms/cycle)	2.4	4.3	8.9
Implementation complexity	Low	Medium	High

capabilities. Model Predictive Control (MPC) was assessed but excluded from direct comparison due to its requirement for accurate predictive models, which are impractical given the complex nonlinear pressure-flow dynamics and unpredictable supply steam pressure fluctuations in this system.

Discussion

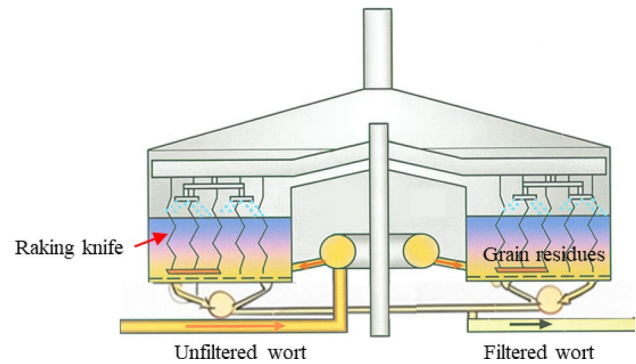
The implementation of the proposed industrial AI framework for steam flow management demonstrates improvements across key performance metrics, validating the systematic approach's effectiveness in addressing complex industrial control challenges.

Quantitative analysis revealed that the AI-enhanced control system achieved a 52.3% reduction in mean overshoot of steam flow rate (from 16.78 to 8.00%), and reductions in mean cumulative steam flow rate deviation. Moreover, mean steam usage deviation decreased from 71.04 to 25.42 kg, representing substantial energy efficiency gains with direct implications for operational costs and carbon emissions.

The adaptive PID controller's pressure-dependent gain scheduling effectively compensated for the nonlinear relationship between steam valve opening and flow rate across varying supply steam pressure conditions. Open-loop testing confirmed the mechanism's effectiveness, with AI control limiting undershoot to 5.7% compared to 18.2% overshoot under non-AI control during setpoint changes. This adaptive behavior addresses the fundamental limitation of fixed-parameter controllers operating under variable conditions.

The systematic five-stage framework proved its practical utility in translating control challenges into implementable solutions. The scenario understanding and problem identification phases successfully identified supply steam pressure as the critical disturbance variable, while the optimization strategy formulation bridged theoretical concepts with practical constraints. The edge computing deployment strategy demonstrated compatibility with existing DCS infrastructure while enabling advanced control capabilities.

The demonstrated energy efficiency gains have particular relevance for industrial sustainability initiatives. The framework's systematic approach addresses common barriers to industrial AI adoption: the theory-practice gap, fragmented implementations, and integration challenges. The successful

**Fig. 16** Lauter tun schematic representation

deployment validates the framework's potential for broader application across diverse industrial control scenarios, establishing a foundation for optimizing process controls through the proposed industrial AI framework.

Case study 2: filtering process optimization

Scenario understanding: filtering process

(1) Process flow investigation

The filtering stage constitutes a critical component in beer production, responsible for separating liquid wort from solid grain residues (spent grain) to provide clarified wort for subsequent boiling operations. This process occurs in lautertuns equipped with specialized mechanical components designed to optimize separation efficiency while maintaining product quality. The filtering stage encompasses four distinct phases: first wort filtration, second wort filtration (sparging), weak wort recovery, and spent grain discharge.

Figure 16 illustrates the schematic representation of a lautertun system used in the filtering process. The vessel features a false bottom with slots that allow liquid wort to pass through while retaining solid grain particles. The raking knife mechanism serves as the primary control element, maintaining optimal grain bed porosity to facilitate efficient filtration. Critical sensors monitor liquid level, raking knife height, turbidity levels, and filtration flow rates across three concentric zones (inner, middle, and outer rings).

Process phase overview: The first wort filtration phase involves filtering high-concentration wort while managing grain bed compaction through raking knife operations. The sparge phase adds water while continuing filtration to obtain medium-concentration wort. The weak wort recovery phase collects residual low-concentration wort, and the spent grain discharge phase removes grain residues through controlled

raking knife movement. Current operations rely on empirical recipe parameters for all control decisions.

(2) System structure clarification

The lauter tun system incorporates multiple sensor technologies providing comprehensive process monitoring capabilities. Differential pressure sensors measure pressure differences between the liquid surface and vessel bottom, indicating grain bed resistance. Liquid level sensors track wort volume throughout filtration phases. Raking knife height sensors provide precise positioning feedback for controlled grain bed management. Turbidity sensors monitor wort clarity across the three drainage zones, ensuring product quality compliance. Flow rate transducers measure filtration rates for each concentric zone, enabling precise flow control and distribution analysis.

The control infrastructure utilizes PLCs interfacing with the DCS through standardized industrial communication protocols. Motor current sensors monitor raking knife power consumption, providing insights into grain bed resistance and mechanical load conditions.

(3) Worker operating modes

The filtering process operates under automated control with operator oversight through HMI systems. Operators monitor real-time parameters including differential pressure trends, flow rate distributions across drainage zones, turbidity levels, and raking knife positioning. Manual intervention capabilities allow operators to adjust setpoints and operational parameters based on visual inspection and process knowledge.

(4) Objective specification

The primary optimization objectives for the filtering process include minimizing total filtration time to reduce energy consumption while maintaining product quality standards, and ensuring wort recovery yield. These measurable objectives align with operational efficiency requirements and provide clear targets for subsequent problem identification and control system enhancement phases.

Problem identification

(1) Data collection and preprocessing

Historical operational data were collected from the facility's DCS covering filtration operations across multiple production batches. The dataset contains data collected over a

6-month operational period (October 2024 to March 2025), with measurements recorded at 8–10 s intervals. Primary variables include differential pressure measurements, raking knife height and motor current, turbidity levels across three drainage zones, flow rates for each zone, liquid level measurements, and temporal data for each filtration phase.

(2) System understanding

Analysis of historical operational data revealed significant limitations in the current empirical recipe-based control approach. The existing PLC system operates exclusively on predetermined recipe parameters without adaptive capabilities to respond to real-time process conditions, resulting in suboptimal performance across varying operational scenarios.

(3) Control limitations analysis

Current control strategies rely on fixed differential pressure thresholds and predetermined time schedules for raking knife positioning across all filtration phases. During active filtration (first wort, sparge, and weak wort recovery), the system cannot adequately respond to dynamic pressure variations or evolving pressure trends, limiting filtration efficiency optimization. For spent grain discharge operations, fixed raking knife residence times based on empirical recipes lead to inefficient discharge operations, with potential time waste when grain discharge is complete but operations continue according to predetermined schedules.

(4) Optimization opportunity identification

The comprehensive analysis identified two primary optimization opportunities:

- (a) **Dynamic Raking Knife Control:** Implementing real-time differential pressure and trend-based raking knife positioning to optimize grain bed porosity.
- (b) **Optimized Discharge Operations:** Implementing motor current-based monitoring for precise discharge completion detection.

Optimization strategy formulation

To address the limitations identified in the problem analysis, this section presents adaptive control strategies that replace fixed empirical parameters with dynamic, condition-responsive mechanisms. The optimization approach centers on implementing real-time monitoring-based adaptive control methods to replace time-based empirical threshold controls,

thereby reducing mean filtration time while maintaining filtration quality standards.

Optimization model

The filtering process optimization problem can be formulated as a multi-objective optimization framework encompassing dynamic raking knife positioning. For the filtering process control system, the optimization problem is formulated as:

$$\min_{\theta_h} J = \sum_{k=1}^4 T_k, \quad (22)$$

Subject to quality constraints ($Turbidity_{avg} \leq Turbidity_{max}$, $Y_{recovery} \geq Y_{min}$), operational constraints ($h_{min} \leq h_{knife}(t) \leq h_{max}$), and safety constraints ($\Delta P(t) \leq \Delta P_{max}$), where T_k represents the duration of phase k , and θ_h represent adaptive control parameter sets for height management.

Adaptive control strategy

The adaptive control framework implements two complementary approaches targeting different operational phases of the filtration process. The first approach addresses raking knife control during active filtration phases through differential pressure monitoring, while the second optimizes discharge operations through motor current analysis.

(1) Active filtration phase

During the active filtration phases (first wort, sparge, and weak wort recovery), the control system employs differential pressure monitoring and trend analysis to manage grain bed compaction dynamically. This approach replaces fixed empirical parameters with adaptive control logic that responds to real-time process conditions.

The adaptive control logic operates based on dynamic pressure conditions. When $\Delta P(t) > \Delta P_{threshold}$ and $\Delta \dot{P}(t) > 0$, indicating progressive grain bed compaction, the raking knife height adjustment is determined by:

$$\Delta h_{knife}(t) = \alpha_1 \cdot [\Delta P(t) - \Delta P_{threshold}] + \alpha_2 \cdot \Delta \dot{P}(t), \quad (23)$$

$$h_{knife}(t+1) = h_{knife}(t) + \Delta h_{knife}(t), \quad (24)$$

where $\Delta P_{threshold}$ is the adjustment threshold, and α_1 and α_2 are adaptive parameters optimized for each filtration phase.

(2) Discharge phase

The spent grain discharge phase requires a different control approach due to the distinct operational characteristics and objectives. This phase optimization focuses on utilizing real-time motor current monitoring to replace fixed time-based discharge schedules with condition-responsive control mechanisms. The discharge control logic operates through motor current analysis:

$$\Delta h_{knife}(t) = \gamma_1 \cdot [I_{motor}(t) - I_{threshold}] + \gamma_2 \cdot \dot{I}_{motor}(t), \quad (25)$$

$$h_{knife}(t+1) = h_{knife}(t) - \Delta h_{knife}(t), \quad (26)$$

where $I_{motor}(t)$ represents current motor current measurement, and $\dot{I}_{motor}(t)$ denotes the rate of current change. Here, γ_1 is the motor current response gain controlling the magnitude of height adjustment based on current deviation, and γ_2 is the motor current rate response gain providing predictive control based on current trends.

When $I_{motor}(t) < I_{threshold}$ and $\dot{I}_{motor}(t) < 0$, it indicates that the gap between the raking knife and the residue layer increases. The raking knife depth can be increased to facilitate faster discharge of the residue layer, and the raking knife height is adjusted according to Eqs. (25)–(26).

Discharge completion is determined when motor current and its derivative remain below specified thresholds for a predetermined duration, ensuring efficient operation while preventing motor overload through safety-constrained control limits.

Both adaptive control strategies are implemented within the established edge computing framework, utilizing pre-computed lookup tables refined through historical performance analysis. This approach combines computational efficiency with real-time adaptability, addressing fundamental limitations of fixed empirical control methods while maintaining practical implementation feasibility.

System deployment

The deployment architecture for the filtering process optimization follows the same edge computing-based hierarchical control framework established in section “[System deployment](#)”. Edge computing devices were positioned within the control network to interface with existing PLCs and DCS infrastructure, providing computational capabilities for advanced control algorithm execution.

The implementation architecture maintains the established communication pathways: sensors and actuators connected to lauter tun equipment communicate with dedicated PLCs via MQTT protocol, PLCs aggregate sensor data and transmit consolidated information to the centralized DCS, and edge computing devices receive operational data

from the DCS while executing adaptive control algorithms locally.

For the filtering process, the edge computing layer implements four specialized control modules corresponding to each filtration phase: first wort filtration control, sparge phase management, weak wort recovery optimization, and spent grain discharge control. These modules continuously monitor process variables including differential pressure, turbidity levels, flow rates, raking knife position, and motor current, generating optimized control signals transmitted through the reverse communication pathway to execute precise control actions.

System commissioning procedures included validation of sensor calibration, verification of control algorithm performance under varying process conditions, and integration testing to ensure seamless operation with existing brewing process management systems.

Verification and evaluation

Performance metrics

System performance evaluation utilized five quantitative metrics designed to assess filtration efficiency, product quality, and resource utilization:

- (1) Total filtration time (T_{total}): measures the cumulative time required for complete filtration across all phases.

$$T_{total} = T_{first} + T_{sparge} + T_{recovery}, \quad (27)$$

- (2) Wort Recovery Yield ($Y_{recovery}$): Quantifies the percentage of recoverable wort successfully extracted during the filtration process.

$$Y_{recovery} = \frac{V_{recovered}}{V_{theoretical}} \times 100\%, \quad (28)$$

where $V_{recovered}$ represents actual recovered wort volume and $V_{theoretical}$ represents theoretical maximum recoverable volume.

- (3) Spent grain discharge time ($T_{discharge}$): quantifies the duration required for complete grain removal.
- (4) Mean turbidity ($Turbidity_{avg}$): measures wort clarity to ensure product quality standards.

Closed-loop testing

Comprehensive performance analysis was conducted using production data from 45 batches, comprising 23 batches under AI-enhanced control and 22 batches under conventional non-AI control. The experimental results (Fig. 17)

demonstrate significant improvements across multiple parameters:

- (1) *Filtration time reduction*: AI control demonstrated substantial time savings with total filtration time reduced by 10.01 min on mean (from 187.52 to 177.51 min), representing a 5.3% improvement and contributing to overall production cycle efficiency improvements.
- (2) *Recovery yield performance*: AI-controlled batches achieved a mean recovery yield of 90.59%, compared to 88.00% under non-AI control, representing a 2.94% improvement in wort recovery efficiency. This enhancement directly translates to increased raw material utilization and improved production economics.
- (3) *Discharge time optimization*: Spent grain discharge operations under AI control averaged 11.27 min compared to 13.35 min under conventional control, representing a 2.08-minute reduction. This improvement results from dynamic motor current monitoring enabling precise discharge completion detection.
- (4) *Product quality maintenance*: Mean turbidity under AI control 29.42 EBC remained lower than non-AI control 20.26 EBC, confirming that operational efficiency improvements did not compromise product quality standards.
- (5) *Energy and cost savings*: The documented operational time reductions provide measurable energy efficiency improvements. Combined filtration and discharge time savings averaging 12.09 min per batch represent a 6.0% reduction in active processing time for these operations. Based on typical industrial brewing equipment specifications with an estimated combined power consumption of 100 ± 50 kW for filtration and discharge systems, the operational time reduction of 201.5 h annually (for 1000 batches) translates to approximately 20,150 kWh of electrical energy savings. This corresponds to an estimated annual CO₂ emission reduction of 10–30 tons depending on regional electricity grid carbon intensity factors, demonstrating quantifiable environmental sustainability benefits alongside operational cost savings.

Discussion

The implementation of adaptive control strategies in filtering process optimization demonstrates significant improvements in operational efficiency while maintaining product quality standards. The systematic replacement of fixed empirical parameters with dynamic control mechanisms addresses fundamental limitations in conventional brewing process control.

Time reductions across all filtration phases contribute to increased production capacity and reduced energy

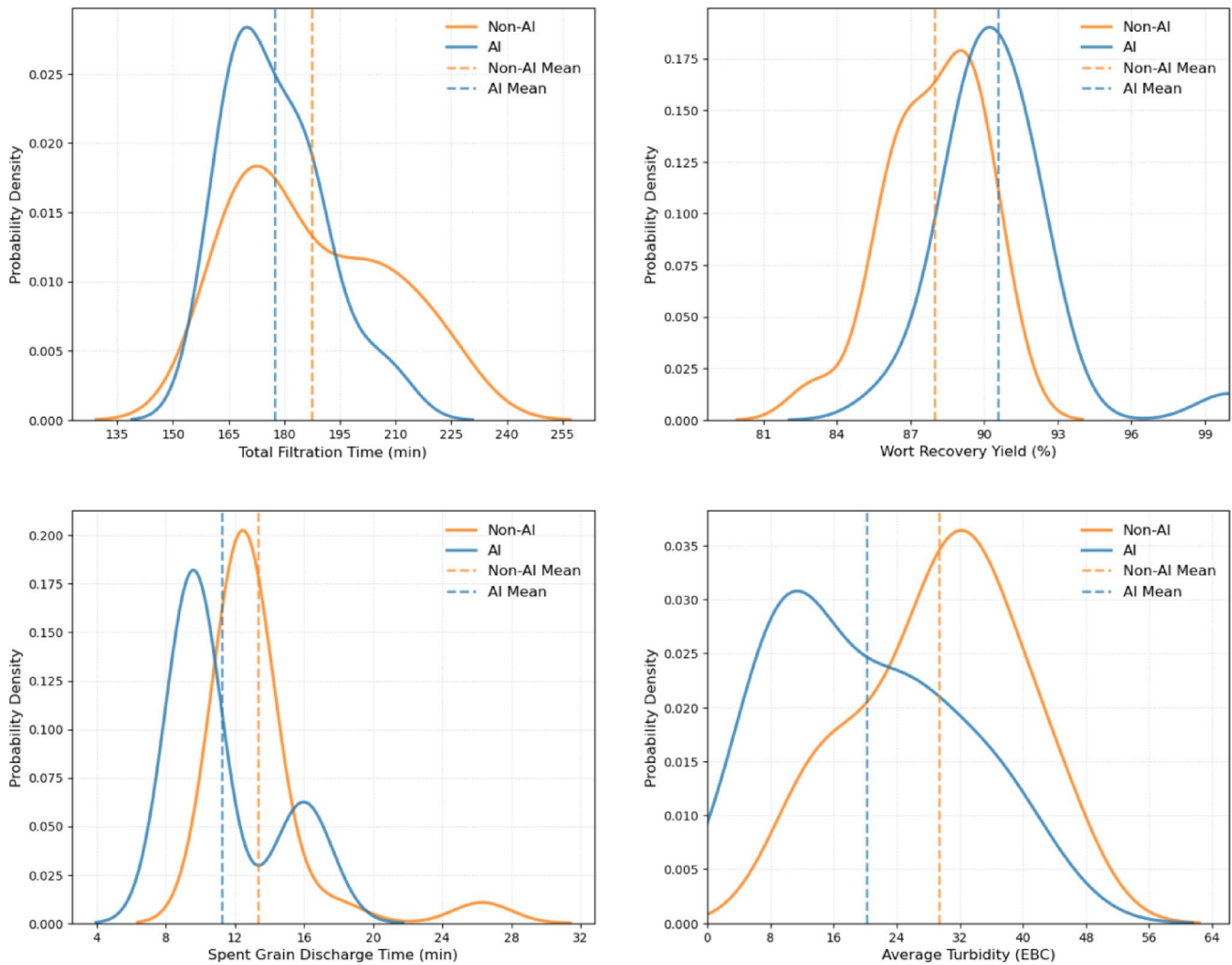


Fig. 17 Performance comparison of non-AI control and AI control in lauter tuns

consumption per unit of product. The successful integration of real-time process monitoring with adaptive control algorithms validates the framework's effectiveness in complex, multi-phase industrial processes. The successful integration with existing control infrastructure demonstrates the framework's compatibility with legacy systems and practical deployment feasibility.

These results establish a foundation for the broader application of the industrial AI framework across diverse brewing operations and related process industries, confirming the systematic approach's effectiveness in translating theoretical optimization concepts into measurable operational improvements.

Discussion

Framework generalizability and applicability

The framework's effectiveness is demonstrated through successful implementation in two distinct process control scenarios: steam flow management and filtering process optimization within the same brewery facility. These applications illustrate the framework's adaptability across different unit operations—from continuous thermal regulation to discrete separation processes—demonstrating its modular design and process-agnostic methodology. The consistent five-stage approach addresses varied control challenges without requiring fundamental structural modifications, validating the framework's transferability across diverse industrial control scenarios.

- (1) *Industrial process applicability*: Building upon this demonstrated versatility, the framework suits various

process categories including continuous control (chemical reactors, distillation columns), batch optimization (pharmaceutical fermentation, food processing), and mixed systems (semiconductor manufacturing, petrochemical complexes). The five-stage methodology systematically identifies control variables and optimization targets across these diverse applications, providing a standardized approach regardless of process type.

- (2) *Cross-Industry application potential*: This process-agnostic foundation enables broader multi-sector applications beyond the brewing industry. Chemical manufacturing opportunities include reactor control and separation optimization, while pharmaceutical applications encompass fermentation control and purification workflows requiring regulatory compliance. Similarly, food processing benefits include temperature control and mixing optimization, and metal processing involves furnace control and energy optimization, all leveraging the same methodological framework.
- (3) *Framework limitations*: However, successful cross-industry implementation faces several constraints that must be acknowledged. Data availability limitations include insufficient sensor density or historical records, while system complexity challenges arise with highly coupled multi-variable processes exceeding current modeling capabilities. Additionally, safety-critical applications require additional fail-safe mechanisms and regulatory compliance protocols beyond the framework's current scope.
- (4) *Implementation adaptations*: To address these limitations and ensure successful deployment, systematic component adaptation becomes essential. Objective functions must reflect industry-specific priorities (energy efficiency, product purity, throughput), while constraint definitions require industry-specific safety margins and regulatory boundaries. Furthermore, control algorithms may need modification based on process characteristics and industry preferences for conservative versus aggressive optimization approaches, ensuring the framework's core methodology adapts appropriately to sector-specific requirements.

Methodological limitations

Despite demonstrated effectiveness, the framework faces several inherent limitations that warrant consideration. Hardware dependencies include requirements for reliable edge computing infrastructure, high-quality sensors, and stable communication networks, which may present cost barriers for resource-constrained facilities. Dynamic system constraints emerge in highly variable environments where rapid process changes may exceed the framework's

adaptation capabilities, particularly when control update frequencies cannot match system dynamics. Data requirements represent another limitation, as the framework's performance depends on sufficient historical data quality and completeness for effective parameter optimization. New equipment installations or significant process modifications may require extended recalibration periods. Implementation complexity demands specialized technical expertise for system integration and ongoing maintenance, potentially limiting adoption in facilities lacking automation capabilities.

However, these limitations also highlight opportunities for future research, including development of more robust adaptation algorithms for highly dynamic systems, simplified deployment procedures for broader accessibility, and enhanced model-free approaches that reduce data dependency requirements. Understanding these constraints enables more informed implementation decisions and guides continuous framework refinement.

Conclusion

This research presents a comprehensive industrial AI framework designed to bridge the gap between theoretical research and practical implementation in process control optimization. The systematic five-stage workflow demonstrates its effectiveness and generalizability through successful applications in both steam flow management and filtration process optimization.

The dual-case empirical validation establishes the framework's versatility across different industrial control challenges. In steam flow management, the framework achieved 52.3% reduction in control overshoot through adaptive parameter adjustment, while in filtration optimization, it delivered 18.6-minute reduction in processing time with improved product quality. These results across fundamentally different control paradigms validate the framework's universal applicability beyond single-domain implementations.

The framework's standardized methodology addresses three critical barriers to industrial AI adoption: the disconnect between theoretical research and practical constraints, fragmented case-specific implementations, and integration challenges with existing infrastructure. The edge computing deployment architecture demonstrates successful integration with legacy DCS systems while maintaining operational reliability and real-time control capabilities.

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Declarations

Conflict of interest Authors declare no Conflict of interest with anyone.

Ethical approval All procedures performed in the work involve no human or animal participants.

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